

Unit III Data Science

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Introduction to Data science - Introduction to Statistics Central Tendency - Data Distribution - Probabilities of Discrete and Continuous Variables- Introduction to Inferential Statistics - Hypothesis Testing - T tests - Chi Squared Tests Correlation. Understanding the types of Predictive Models - ~~Univariate Analysis~~ - ~~Bivariate Analysis~~ - ~~Treating Missing Values~~ - ~~How to treat Outliers~~ - Transforming the Variables - ~~Basics of Model Building~~.

Data Science - Introduction

Data Science is a deep study of the **massive amount of data** by extracting **meaningful insights** from raw, structured, and unstructured data that is processed using the scientific method, different technologies, and algorithms.

We want to **travel** from **station A** to station B by car. Now, we need to make some decisions such as **which route will be the best** route to reach faster at the location, which route there will be no **traffic jams**, and which will be cost-effective. All these decision factors will act as **input data**, and we will get an appropriate answer from these decisions, this analysis of data is called data analysis, which is a part of data science.

Tools for Data Science

Data Analysis tools: R, Python, Stata, Statistical Analytics Software (SAS), Jupyter Note Book, R Studio, MATLAB, Excel, RapidMiner.

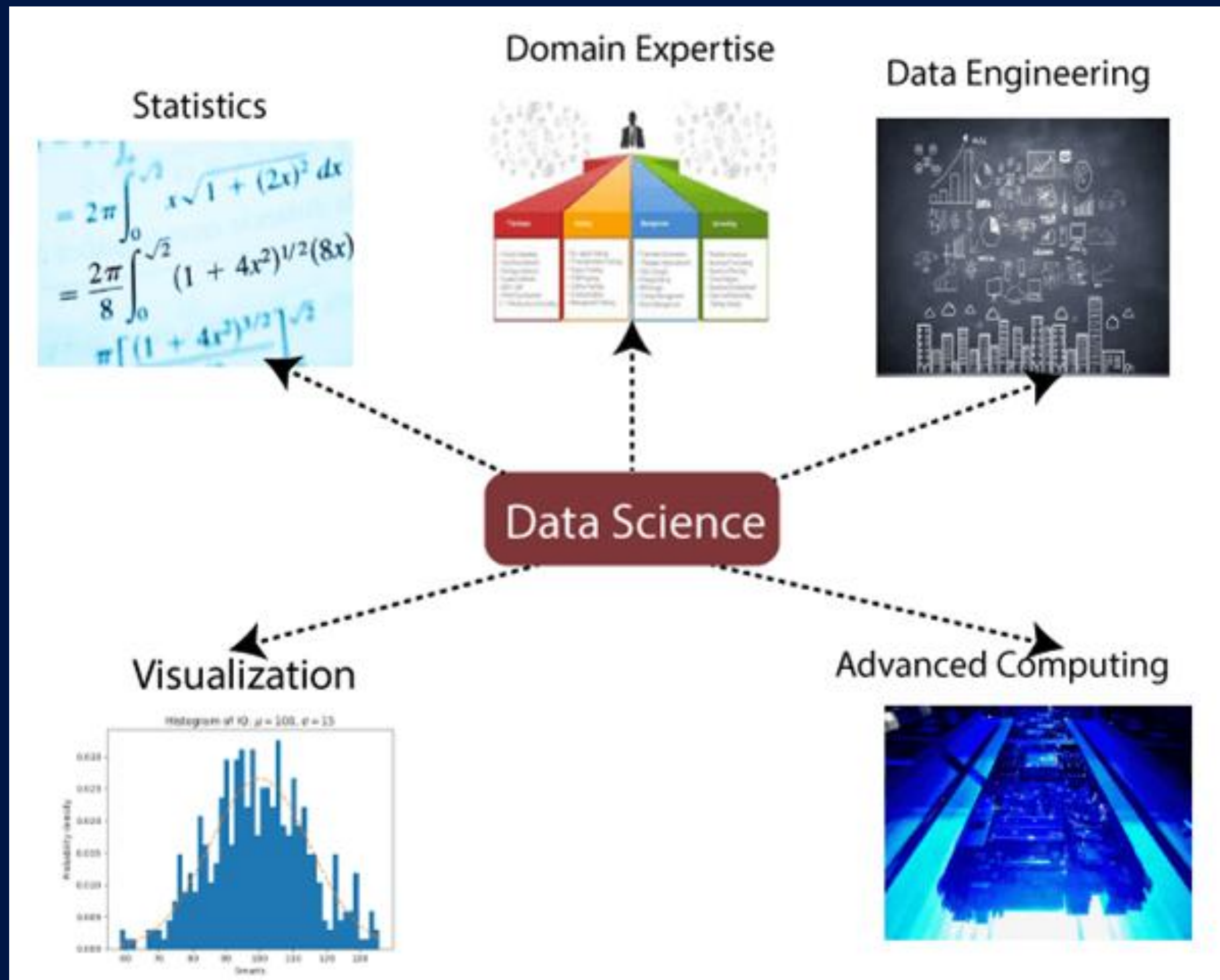
Data Warehousing: ETL, SQL, Hadoop, Informatica/Talend, AWS Redshift

Data Visualization tools: R, Jupyter, Tableau, Cognos.

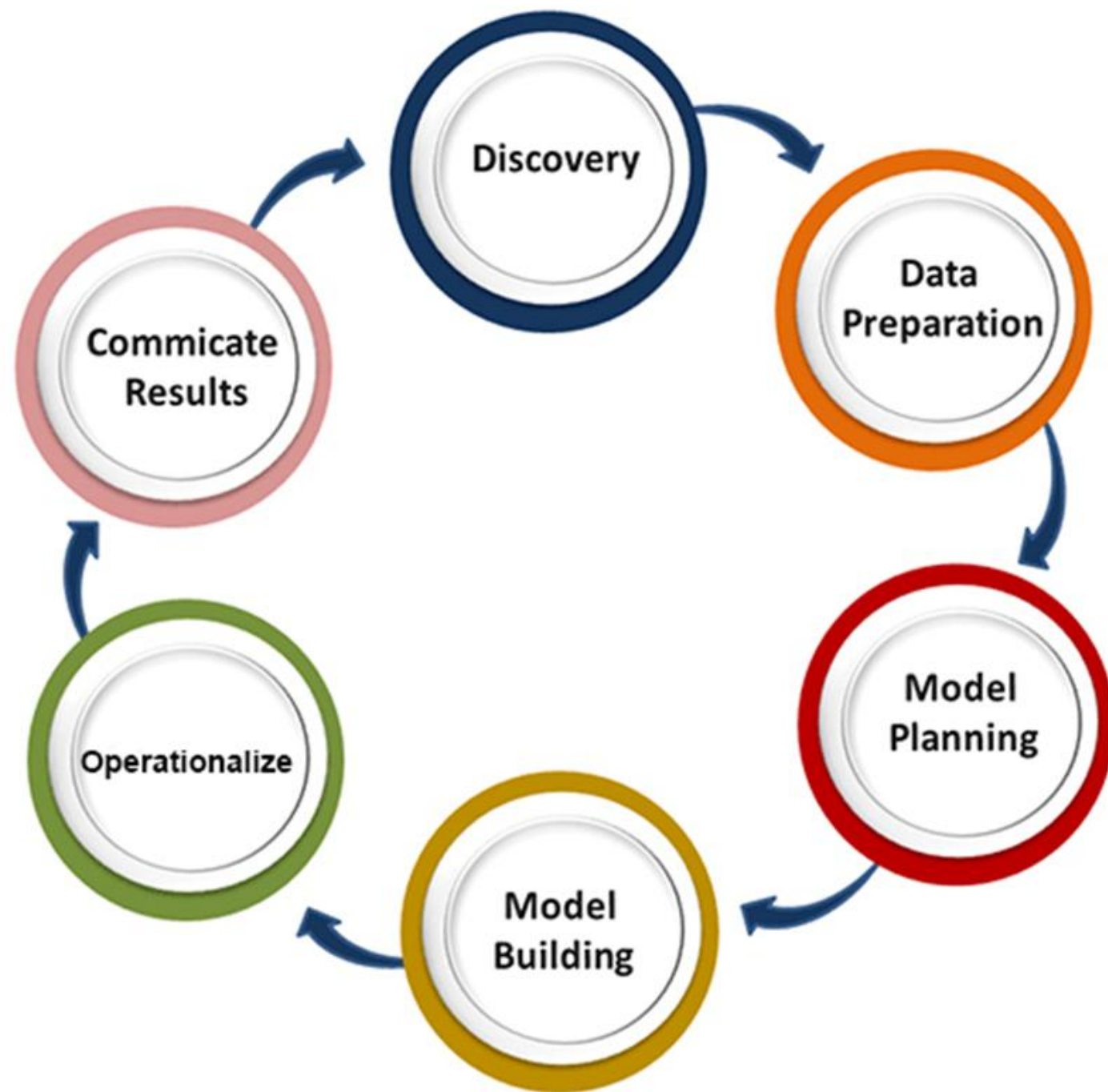
Machine learning tools: Spark, Mahout, Azure ML studio.

Main Components of Data Science

- 1. Statistics:** Statistics is a way to collect and analyze numerical data in a large amount and find meaningful insights from it.
- 2. Domain Expertise:** Domain expertise means specialized knowledge or skills in a particular area. In data science, there are various areas for which we need domain experts.
- 3. Data engineering:** Involves acquiring, transforming, storing, and retrieving the data. Data engineering also includes metadata (data about data) to the data.
- 4. Visualization:** Data visualization is meant to represent data in a visual context so that people can easily understand the significance of data. Data visualization makes it easy to access a huge amount of data in visuals.
- 5. Advanced computing:** Involves designing, writing, debugging, and maintaining the source code of computer programs.
- 6. Mathematics:** Mathematics involves the study of quantity, structure, space, and changes. For a data scientist, knowledge of good mathematics is essential.
- 7. Machine learning:** Machine learning is the backbone of data science. Machine learning is all about providing training to a machine so that it can act as a human brain. In data science, we use various machine learning algorithms to solve the problems.



Data Science Life Cycle



- 1. Discovery:** Involves asking the **right questions**. Determine the basic requirements, priorities, and project budget, number of people, technology, time, data, end goal, and then we can frame the business problem on the first hypothesis level.
- 2. Data preparation:** Data preparation is also known as Data Munging or Data wrangling (Cleaning and Transforming). In this phase, we need to perform the following tasks:
 - ❖ Data cleaning (e.g. removing null data, outliers etc.)
 - ❖ Data Reduction
 - ❖ Data integration (because data are from many sources)
 - ❖ Data transformation (normalization, encoding etc.)

Data Science Life Cycle

3. Model Planning: Determine the various methods and techniques to establish the relation between input variables. We will apply Exploratory data analytics(EDA) by using various statistical formulas and visualization tools to understand the relations between variables and to see what data can inform us. Common tools used for model planning are:

- ❖ SQL Analysis Services
- ❖ R
- ❖ SAS
- ❖ Python

4. Model-building: Create datasets for training and testing purposes. We will apply different techniques such as association, classification, and clustering, to build the model.

Following are some common Model building tools:

- SAS Enterprise Miner (Data Mining Software)
- WEKA
- SPCS (Statistical Process Control Software) Modeler
- MATLAB

5. Operationalize: Deliver the final reports of the project, along with briefings, code, and technical documents. Provides a clear overview of the complete project performance and other components on a small scale before the full deployment.

6. Communicate results: Check if we reach the goal, which we have set in the initial phase. We will communicate the findings and final results with the business team.

Applications of Data Science

i) Image recognition and speech recognition:

When you upload an image on Facebook and start to tag your friends. This uses an image recognition algorithm, which is part of data science.

When you say something using, "Ok Google, Siri, Cortana", etc., and these devices respond as per voice control, this is possible with a speech recognition algorithm.

ii) Gaming world:

In the gaming world, the use of Machine learning algorithms is increasing day by day. EA Sports, Sony, and Nintendo, are widely using data science for enhancing user experience.

iii) Internet search:

All the search engines Google, Yahoo, Bing, Ask, etc. use data science technology to make the search experience better, and you can get a search result in a fraction of a seconds.

iv) Transport:

Transport industries also use data science technology to create self-driving cars. With self-driving cars, it will be easy to reduce the number of road accidents.

Applications of Data Science Contd. ...

v) Healthcare:

Data science is being used for tumor detection, drug discovery, medical image analysis, virtual medical bots, etc.

vi) Recommendation systems:

Most companies, such as Amazon, Netflix, Google Play, etc., are using data science technology to make a better user experience with personalized recommendations. Such as, when you search for something on Amazon, and you start getting suggestions for similar products, this is because of data science technology.

vii) Risk detection:

Finance industries always had an issue of fraud and risk of losses, but with the help of data science, this can be rescued. Most finance companies are looking for data scientists to avoid risk and any type of losses with an increase in customer satisfaction.

Variables – Categories & Identification

Dataset to know whether a student will play cricket or not.

Student_ID	Gender	Prev_Exam_Marks	Height (cm)	Weight Caregory (kgs)	Play Cricket
S001	M	65	178	61	1
S002	F	75	174	56	0
S003	M	45	163	62	1
S004	M	57	175	70	0
S005	F	59	162	67	0

Type of Variable

Data Type

Variable Category

Predictor Variable

- Gender
- Prev_Exam_Marks
- Height
- Weight

Target Variable

- Play Cricket

Character

- StudentID
- Gender

Numeric

- Play Cricket
- Prev_Exam_Marks
- Height
- Weight

Categorical (Qualitative Or Nominal)

- Gender
- Play Cricket

Continuous (Quantitative) (Numeric)

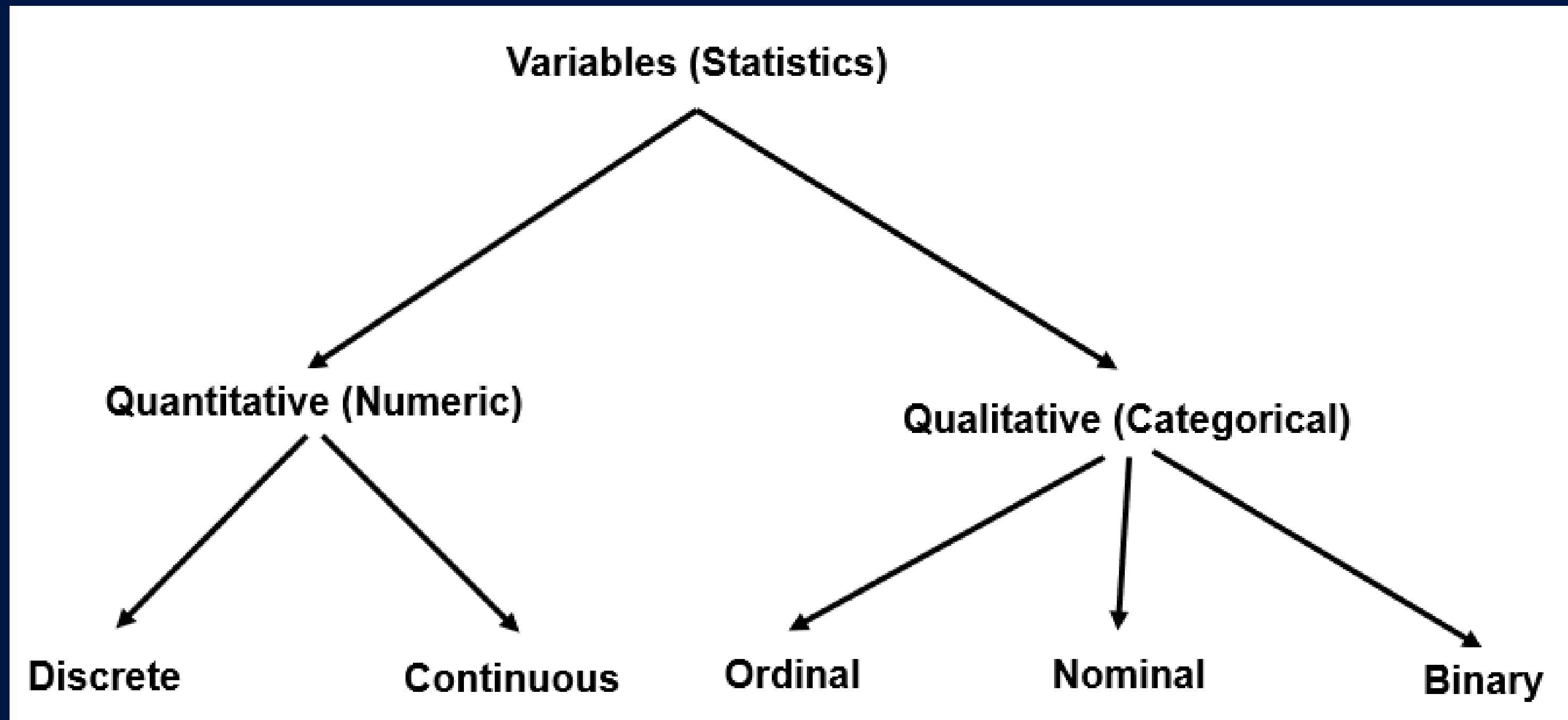
- Prev_Exam_Marks
 - Height
 - Weight
- (Can also be Discrete)

Identify type of variable i.e. whether **predictor** (**input**) variables, or **target** (**Output**) variables.

Identify the **data type** of variables

Identify the **category** of variables.

Categories of Variables



Quantitative Variables (Numeric): Measurable quantity.

These variables are names and can have only numerical values.

Examples: **Number of students** in a class, **Age** of an individual, **Height** of an individual

Quantitative variables can further be classified as discrete or continuous.

a) Discrete variable:

Discrete variables are **numeric** variables that have a **countable number of values between** any two values. A discrete variable is always numeric. For example, the number of **customer complaints** or the number of flaws or defects.

b) Continuous variable:

Continuous variables are **numeric** variables that have an **infinite number of values between** any two values. A continuous variable can be numeric or **date/time**. For example, the **length of a part** or the date and time a payment is received.

We can use many different metrics to **summarize quantitative** variables, including:

Measures of **central tendency** like the mean, median, and mode.

Measures of **dispersion** like the range, interquartile range, and standard deviation.

Qualitative Variables (Categorical):

These variables are **names** and can have **categories or labels** as values. These variables describe **attributes or qualities**.

Categorical variables contain a **finite number of categories or groups**.

Categorical data **might not have a logical order**.

Examples:

Gender (e.g. “male”, “female”)

Level of education (e.g. “high school”, “UG”, “PG”)

Eye color (e.g. “blue”, “green”, “brown”)

Student_ID	Gender	Prev_Exam_Marks	Height (cm)	Weight Caregory (kgs)	Play Cricket
S001	M	65	178	61	1
S002	F	75	174	56	0
S003	M	45	163	62	1
S004	M	57	175	70	0
S005	F	59	162	67	0

We can only use **frequency tables** and **relative frequency** tables as **metrics** to summarize qualitative variables.

There are three types of qualitative variables.

- **Ordinal:** represents data with an order (e.g. **rankings**).
- **Nominal:** represents group names (e.g. brands or species names).
- **Binary:** represents data with a yes/no or 1/0 outcome (e.g. win or lose).

Ordinal data is classified into categories within a variable. However, the distances between the categories are uneven or unknown.

For example, the variable “frequency of physical exercise” can be categorized into the following:

1. Never	2. Rarely	3. Sometimes	4. Often	5. Always
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There is a **clear order** to these categories, but we cannot say that **the difference** between “**Rarely**” and “**Some Times**” is exactly the same as that between “**Often**” and “**Always**”. Therefore, this scale is ordinal.

Nominal data means “in name” so nominal data can only be labelled. It **does not have a rank order**, equal spacing between values, or a true zero value. Nominal data is labelled into mutually exclusive categories within a variable. These categories cannot be ordered in a meaningful way. For example, preferred **mode of transportation** is a nominal variable, because the data is sorted into categories: **car, bus, train, tram, bicycle**, etc.

If the predictor (input) has many levels, then it may be best to treat it as a continuous variable. Treating a predictor as a continuous variable implies that a simple linear or polynomial function can adequately describe the relationship between the response (output) and the predictor.

When you treat a predictor as a categorical variable, a distinct response value is fit to each level of the variable without regard to the order of the predictor levels.

Identify the Type of the Variable

Variables – Categories & Identification

Dataset to know whether a student will play cricket or not.

Student_ID	Gender	Prev_Exam_Marks	Height (cm)	Weight Caregory (kgs)	Play Cricket
S001	M	65	178	61	1
S002	F	75	174	56	0
S003	M	45	163	62	1
S004	M	57	175	70	0
S005	F	59	162	67	0

Identify type of variable i.e. whether **predictor** (**input**) variables, or **target** (**Output**) variables – Play Cricket is the output variable and all others are input variables.

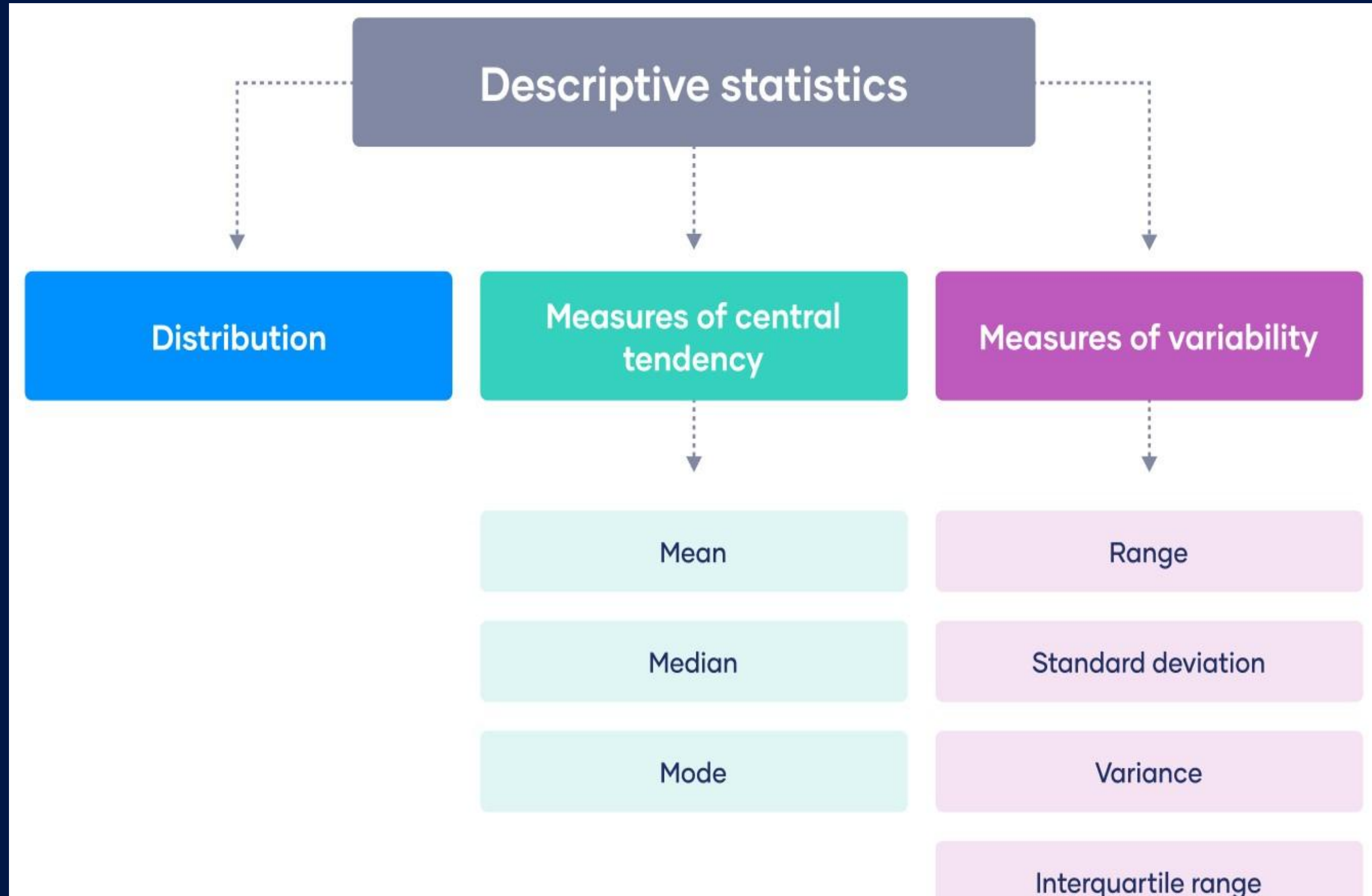
Identify the **data type** of variables – ID, Gender – Strings
Marks – Integer
Height, Weight – Float
Play Cricket – Logical (Yes or No)

Identify the **category** of variables. ID-Nominal (Qualitative) - purpose is identification not ranking(i.e. not ordinal)
Gender – Categorical (Qualitative)
Height – Continuous (Quantitative)
Weight – Continuous (Quantitative)
Play Cricket – Binary (Qualitative) (Yes/No)

If "Weight Category" refers to the specific numerical weight of the student, it is Quantitative. If it refers to a standardized weight class represented by a number, it is Qualitative.

Descriptive Statistics & Measures of Central Tendency

Descriptive statistics allow **describe the data** set, while **inferential statistics** allow to make **inferences** (or predictions) based on the data set.



Distribution:

Distribution refers to how data values are spread across possible values, with frequency indicating how often each value appears.

A **data set** is a collection of responses or **observations** from a **sample or entire population**.

a) Normal distribution

Data is **symmetrically distributed** with no skew. Most values cluster around a **central region**, with values **tapering off** as they go further away from the center.

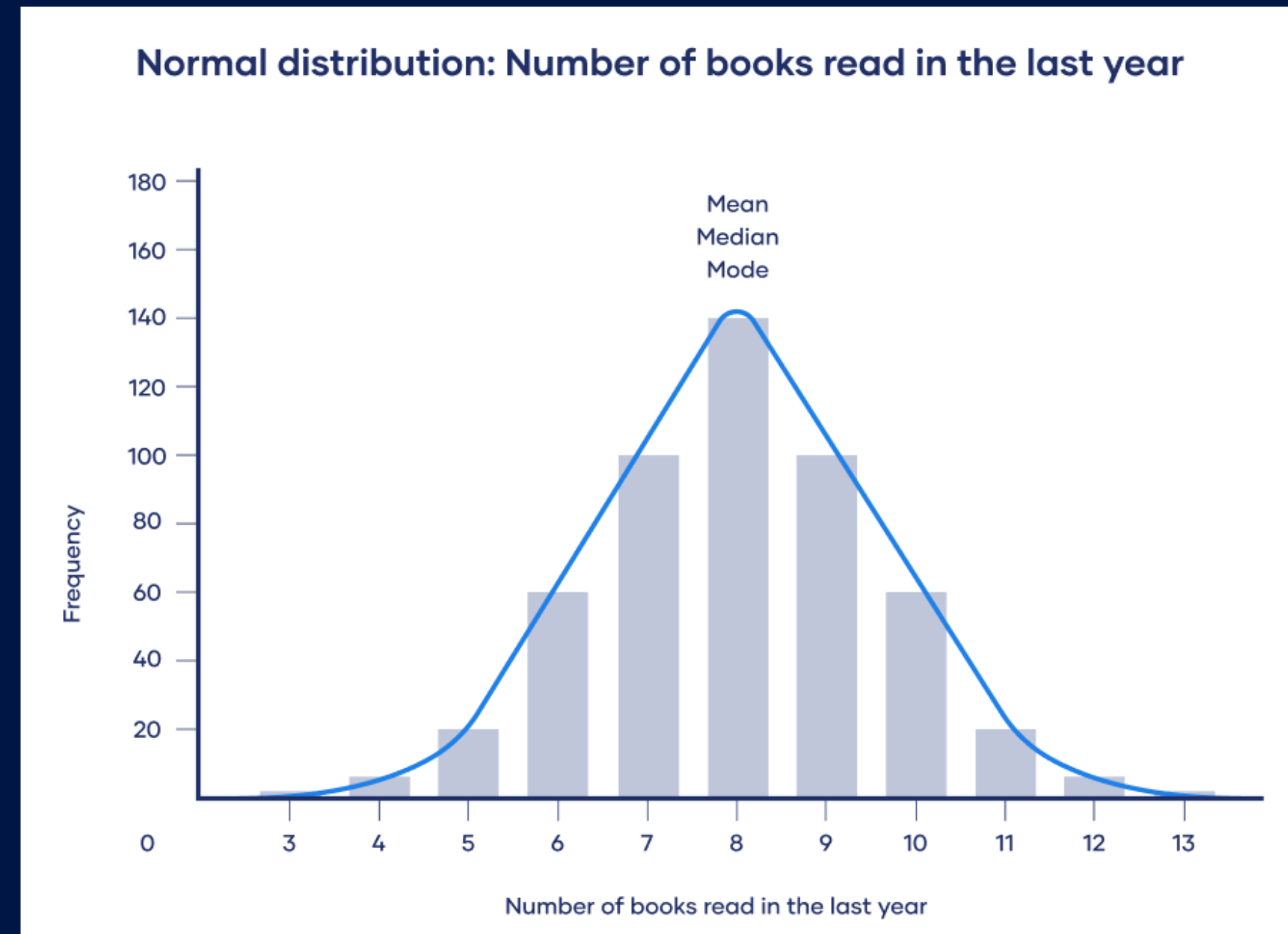
consider the histogram shown here. It gives the **number of persons read (frequency) the books vs the number of books read**. It is a bell-shaped curve. So, it follows the normal distribution.

8 books were read by 140 persons

X-axis → Number of books studied

Y-axis → Number of persons

For a normal distribution,
mean= mode=median.



Normal distribution (Gaussian distribution or bell curve)

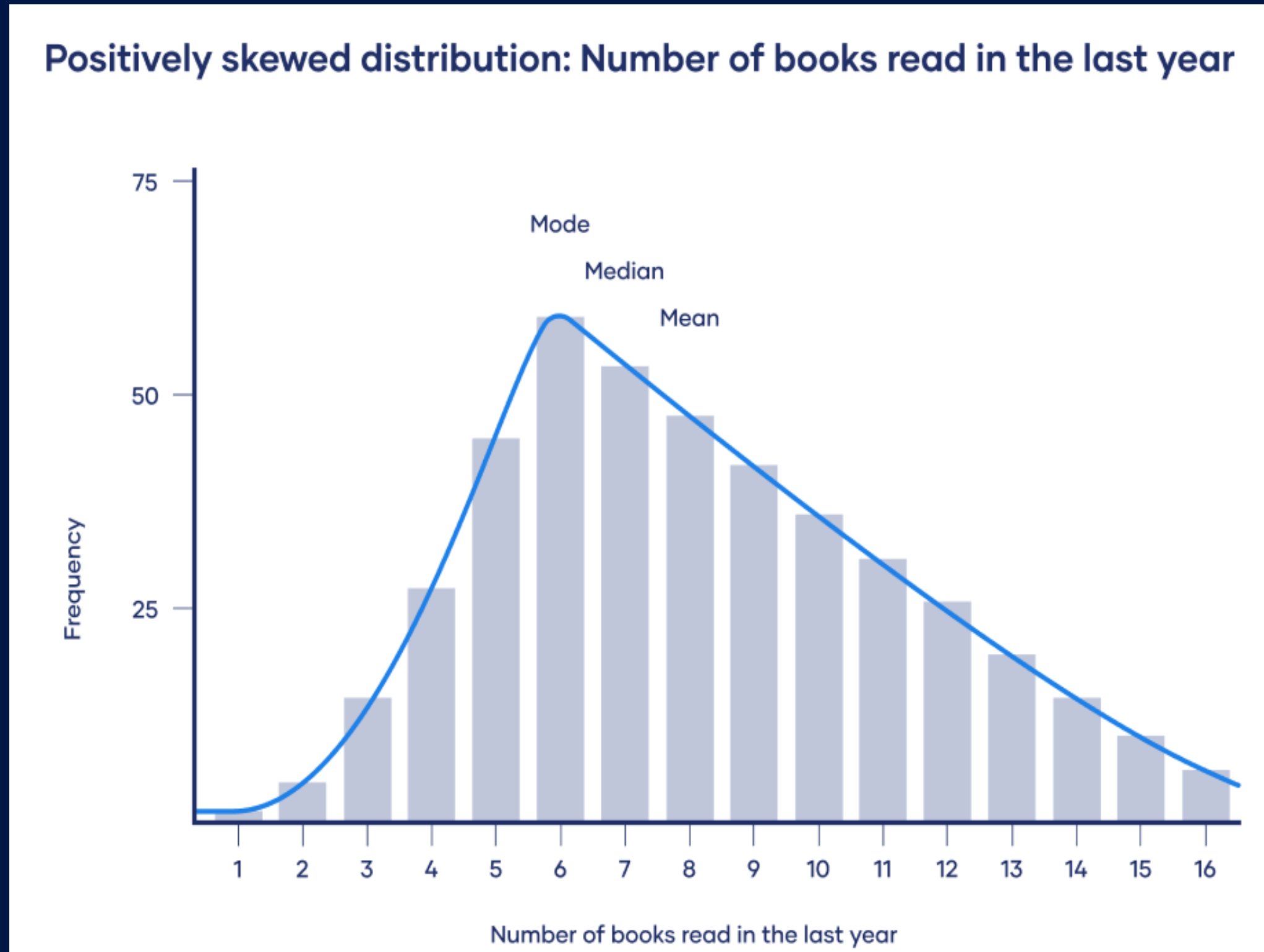
b) Skewed distribution

One side from the centre has a more spread-out and longer tail with **fewer scores (frequency)** at one end than the other.

In the histogram given below, the distribution is skewed to the right because the direction of the tail is at the right.

It is also known as a positively skewed distribution and the central tendency (mean) of the dataset is on the lower end of possible scores

In a positively skewed distribution,
mode < median < mean.



Measure of Central Tendency:

a) Mean

The arithmetic mean of a dataset (which is different from the geometric mean) is the sum of all values divided by the total number of values. Here all values are used.

Participant	1	2	3	4	5
Reaction time (milliseconds)	287	345	365	298	380

First, you add up the sum of all values:

$$\sum x = 287 + 345 + 365 + 298 + 380 = 1\,675$$

Then you calculate the mean using the formula

$$\frac{\sum x}{n}$$

There are 5 values in the dataset, so $n = 5$ and the mean is

$$\bar{x} = \frac{1\,675}{5} = 335$$

Mean ($\bar{x}$): 335 milliseconds

b) Median

The median of a dataset is the value that is exactly in the middle when it is ordered from low to high. If the data has categorical values as given below,

Participant	1	2	3	4	5	6	7
Speed	Medium	Slow	Fast	Fast	Medium	Fast	Slow

to find the median, you first order all values from low to high. Then, you find the value in the middle of the ordered dataset

Ordered dataset	Slow	Slow	Medium	Medium	Fast	Fast	Fast
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Ans: Median: Medium

If there are even number of data, order them from low to high and take the average of the middle two numbers. For example, find the median of $[45, 47, 52, 52, 53, 55, 56, 58, 62, 80]$.

1. Arrange the data from lower value to higher value.
2. Since there are even number of data, there is no single middle value, so, take the average of the middle two number i.e. $(53+55)/2 = 54$

c) Mode

The mode is the **most frequently occurring** value in the dataset.

The mode is most applicable to **nominal** data.

For **continuous or ratio variables**, the mode may not be a helpful measure

To find the mode, sort the dataset **numerically or categorically** and select the response that occurs most frequently.

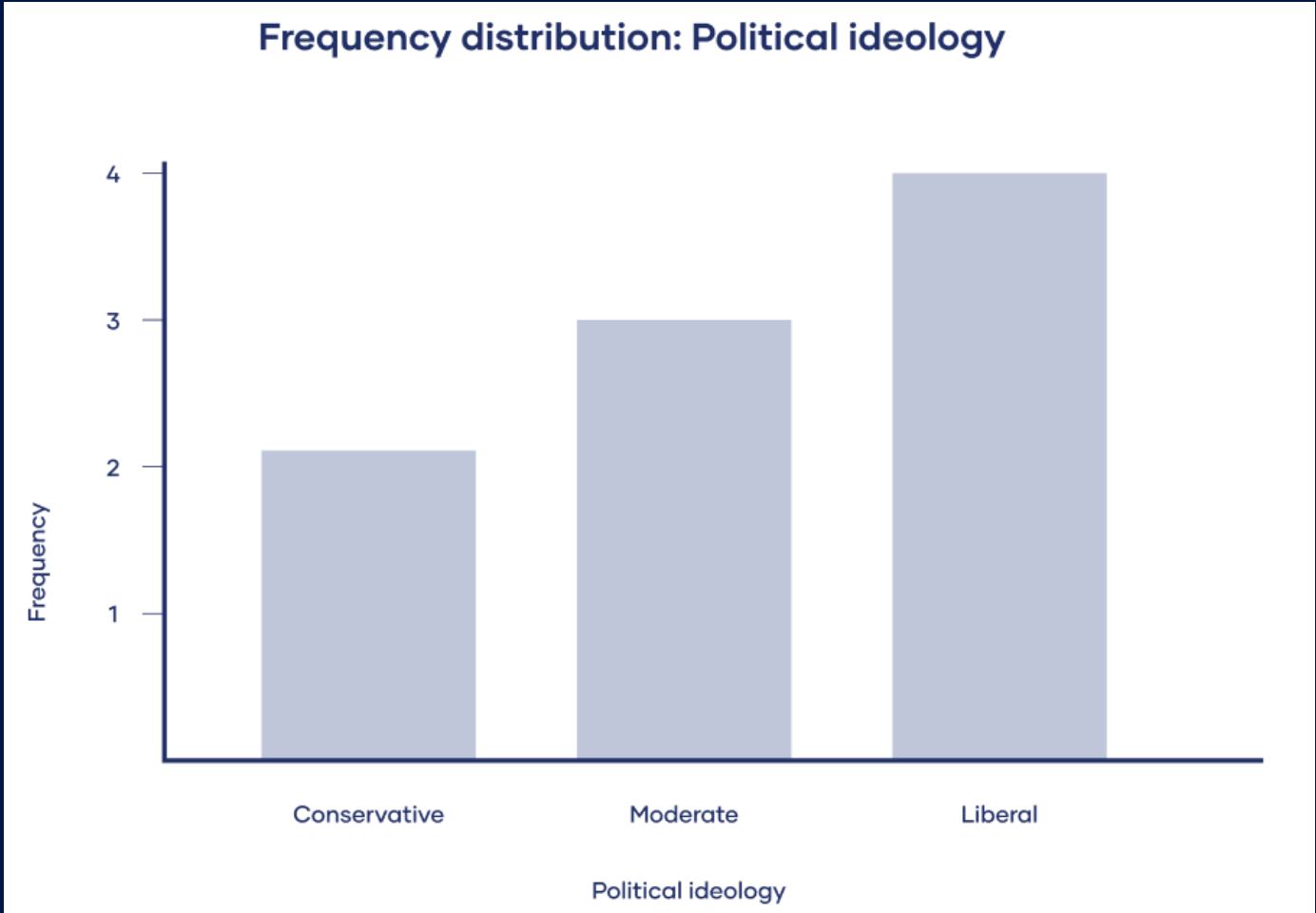
Example: Finding the mode

In a survey, ask 9 participants whether they identify as conservative, moderate, or liberal and record them.

To find the mode, create a frequency table and **sort the data by category** and find which response was chosen most frequently.

If the data occurs **only once**, then there is **no mode**

Political ideology	Frequency
Conservative	2
Moderate	3
Liberal	4



Measure of Variability:

Measures of variability indicates how the response values are **spread out**.

The **range**, **standard deviation** and **variance** each reflect different aspects of spread.

a) Range

The range gives you an idea of **how far apart** the most extreme response scores are.

To find the range, simply subtract the lowest value from the highest value.

Ordered data set: 0, 3, 3, 12, 15, 24

Range: $\text{max} - \text{min} = 24 - 0 = 24$

b) Standard Deviation (σ)

The standard deviation (s or SD) is the **average amount of variability** in the dataset.

It tells you, on average, how far each score lies from the mean. The larger the standard deviation, the more variable the data set is.

There are six steps for finding the standard deviation:

1. List each score (x_i) and find their mean (μ).
2. Subtract the mean from each score to get the deviation from the mean.
3. Square each of these deviations.
4. Add up all of the squared deviations.
5. Divide the sum of the squared deviations by Total number of Data (**N**).
6. Find the square root of the number you found.

$$\sigma = \sqrt{\frac{\sum_{i=1}^N (x_i - \mu)^2}{N}}$$

NB: if divided by N then the std is for entire Population. If divided by N-1 then it is for sample population

Variability Contd. ...

Complete Steps 1 through 4.

Step 5: **Sample variance** $s^2 = 421.5/5 = 84.3$

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

NB: if divided by N then the variance is for entire Population.

Step 6: $\sqrt{84.3} = 9.18$ (standard deviation)

From learning that $s = 9.18$, you can say that on average, each score deviates from the mean by 9.18 points.

Raw data	Deviation from mean	Squared deviation
15	$15 - 9.5 = 5.5$	30.25
3	$3 - 9.5 = -6.5$	42.25
12	$12 - 9.5 = 2.5$	6.25
0	$0 - 9.5 = -9.5$	90.25
24	$24 - 9.5 = 14.5$	210.25
3	$3 - 9.5 = -6.5$	42.25
M = 9.5	Sum = 0	Sum of squares = 421.5

c) Sample Variance

The variance is the **average of squared deviations from the mean**. Variance reflects the degree of spread in the data set. The larger the variance the more spread of the data.

To find the variance, simply square the standard deviation.

In the above example,

$$s = 9.18$$

$$\text{So, Sample Variance} = s^2 = 84.3$$

d) Inter Quartile Range (IQR)

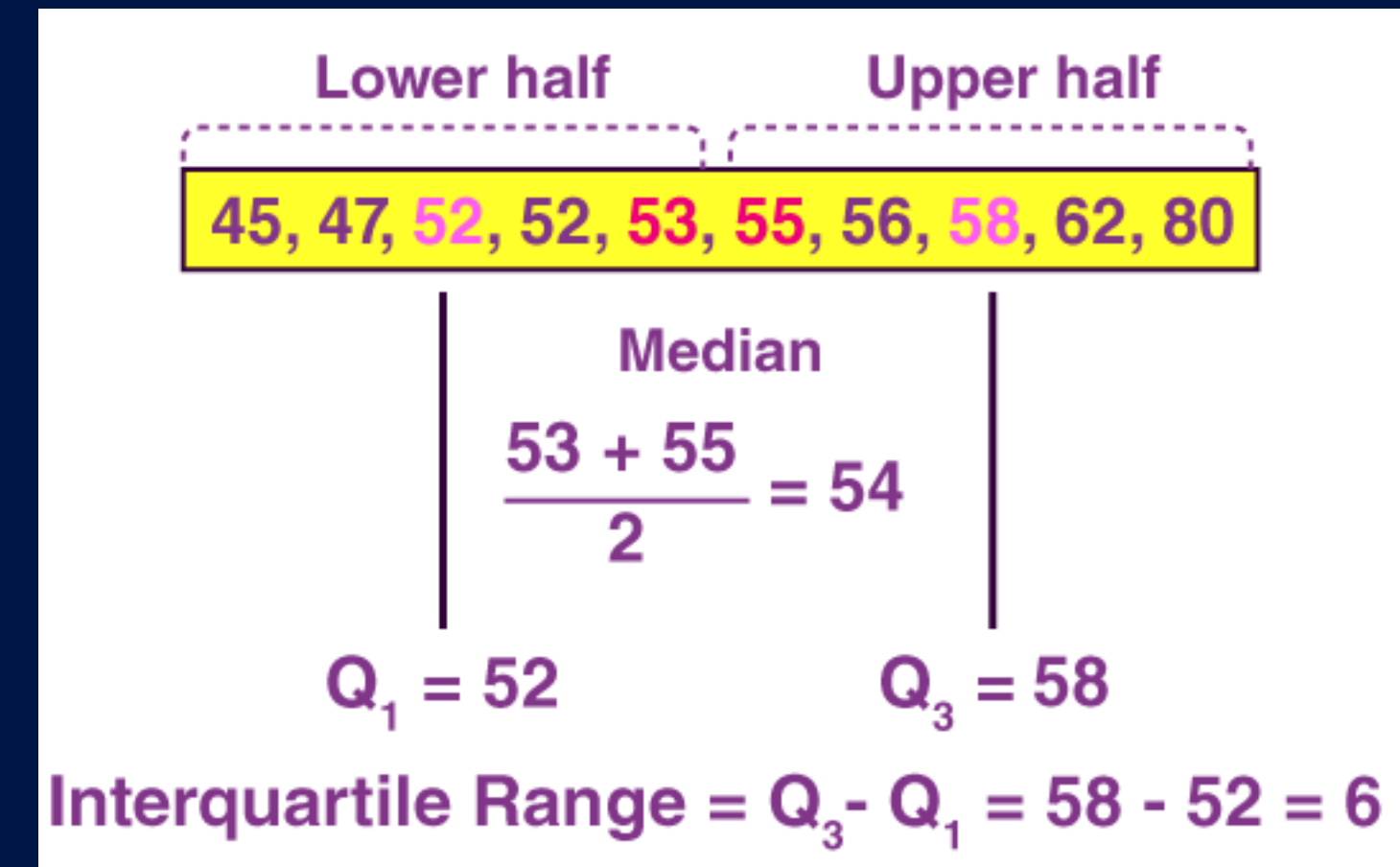
The interquartile range (IQR) is the range of values that resides in the middle of the scores. It is a measure of the dispersion of the data. When a distribution is skewed, and the median is used instead of the mean to show a central tendency, the appropriate measure of variability is the Interquartile range. The interquartile range defines the difference between the third and the first quartile. Quartiles are the partitioned values that divide the whole series into 4 equal parts. So, there are 3 quartiles.

The First Quartile is denoted by **Q1** known as the **lower quartile**

The second Quartile is denoted by **Q2** and it is the **median** of the data set

The third Quartile is denoted by **Q3** known as the **upper quartile**.

Therefore, the interquartile range (**IQR**) is equal to the upper quartile minus the lower quartile.



$$\text{IQR} = \text{Upper Quartile} - \text{Lower Quartile} = Q_3 - Q_1$$

Inferential Statistics

Inferential statistics help you come to conclusions and **make predictions** based on the data.

Mostly you acquire data from samples instead of from the whole population. So, you can use inferential statistics to **understand the larger population** from which the **sample** is taken.

With inferential statistics, it's important to use **random and unbiased sampling** methods.

If your sample isn't representative of your population, then you can't make valid statistical inferences or generalize.

Since the size of a sample is always smaller than the size of the population, some of the population isn't captured by sample data.

This creates **sampling error**, which is the difference between the true population values (called parameters) and the measured sample values (called statistics).

Inferential Statistics Contd. ...

There are two important types of estimates you can make about the population:

i) Point estimate: A single value estimate of a parameter. For instance, a sample mean is a point estimate of a population mean.

ii) Interval estimate: Gives you a range of values where the parameter is expected to lie. A confidence interval is the most common type of interval estimate. A 95% confidence interval means that if you repeat your study with a new sample in exactly the same way 100 times, you can expect your estimate to lie within the specified range of values 95 times. Confidence intervals take sampling error into account.

Hypothesis Testing

A hypothesis is an “educated guess” about a specific parameter or population.

Hypothesis testing is done to support the certainty (probability of success) of the that guess or findings

In hypothesis testing, two mutually exclusive statements about a parameter or population are evaluated to decide which statement is best supported by sample data.

There are two types of hypothesis testing

i) Parametric testing

ii) Non-parametric Testing

Population (whole Dataset) → Sample (random, unbiased portion of the Data)

Output – Prediction – Inference

hypothesis – Educated guess on the population

Hypothesis test – To confirm the success (probability) of the guess made.

Parametric – e.g. height, population should be normal distribution.

Hypothesis Testing

1. Parametric testing:

Hypothesis tests including a **specific parameter** (e.g. height) are called parametric tests. In parametric tests, the population is assumed to have a **normal distribution** (e.g., the height of people in a class).

A normal distribution, also known as a **Gaussian distribution or bell curve**, is a continuous probability distribution that is symmetric around its mean, which is also its median and mode.

Example 1:

Hypothesis 1: Average order value has increased since last financial year

Parameter: Mean order value

Test type: **one-sample (one group)** parametric test (assuming the order value follows a **normal distribution**)

Example 2:

Hypothesis 2: Investing in A brings a **higher return** than investing in B

Parameter: **Difference** in mean return

Test type: **two-sample (two groups)** parametric test, also AB test (assuming the return follows a **normal distribution**)

Hypothesis Testing

In **parametric statistical tests**, the data to be used should have the following characteristics:

- 1. Independence of observations (no autocorrelation):** The observations or variables included in the test are not related.
- 2. Homogeneity of variance:** The **variance** within each group being compared is **similar among all groups**. If one group has much more variation than others, it will limit the test's effectiveness.
- 3. Normality of data:** The data follows a **normal distribution (bell curve)**. This assumption applies only to quantitative data.

Parametric tests can't test hypotheses about the distribution of a **categorical variable**, but they can involve a categorical variable as an independent variable

The most common types of **parametric tests** include **comparison tests**, **correlation tests**, and **regression tests**.

If the data available for testing do not meet the assumptions of **normality** or **homogeneity of variance**, a **nonparametric statistical test** may be performed, which allows to make comparisons without any assumptions about the data distribution.

Hypothesis Testing

2. Non-parametric testing:

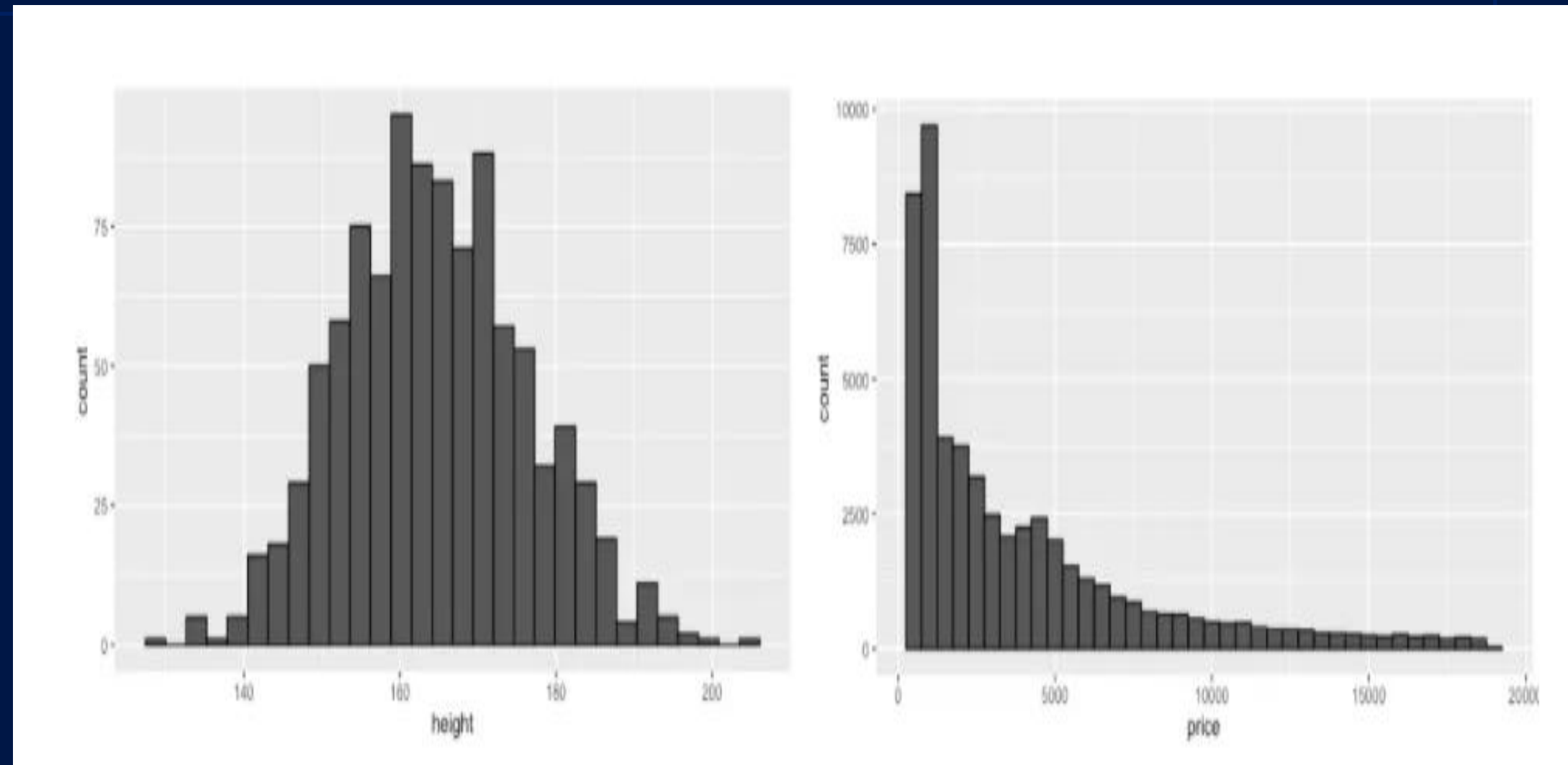
Non-parametric tests (also distribution-free tests) are used when the parameters of a population **cannot be assumed to be normally distributed**. Non-parametric doesn't mean that you do not know anything about a population but rather that it is not normally distributed. For example, the **price of diamonds** seems **exponentially distributed**.

Example 3:

Hypothesis 3: The new user interface converts **more** users into **customers** than the expected 30%

Parameter: none

Test type: one-sample, non-parametric test (assuming the number of customers is not normally distributed)



Left: example of normally distributed data.

Right: example of non-normal data distribution.

Hypothesis & Testing the Hypothesis

Aim: To select two investments which give different returns

i) Null Hypothesis H_0 or H_0

H_0 : There is **no difference** in return between investment A and B
(OR)

The **difference** in return between investment A and investment B **is zero**

H_0 test: fix a **significance level α (alpha)** (e.g. 0.1, 0.01, 0.05 etc)

if the difference $< \alpha$; **hypothesis is true** - reject the null hypothesis because we want to
select investments with different returns

if the difference $> \alpha$; **hypothesis is false** - **select the null hypothesis**

ii) Alternate Hypothesis H_1 or H_1

H_1 : The difference in return between investment A and B is greater than zero

This means $A > B$ or $B > A$ you have to test on both sides – **Two-tailed test**

H_1 : The return of A is greater than the return of B - **one-tailed test**

Null (H_0) and Alternative (H_1) Hypotheses

More Explanation:

The **null** and **alternative hypotheses** are the two mutually exclusive statements about a parameter or population. The null hypothesis (often abbreviated as H_0 or H_0) claims that there is **no effect** or **no difference**. The alternative hypothesis (often abbreviated as H_1 or H_A or H_1 or H_a) is what **you want to prove**.

Null hypothesis often include phrases such as “**no effect**”, “**no difference**” “**no relationship**” written in mathematical terms.

Example 1:

Hypothesis: Investing in A brings a **higher return** than investing in B

Parameter: Difference in mean return

In this example,

H_0 : The **difference** in the return between A and B is **zero**. (There is no effect i.e. no difference)

H_1 : The **difference** in the return between A and B **> zero**. (There is an effect i.e. difference exists – Alternate hypothesis)

Testing the Null Hypothesis

A **significance level (α)** (is used to test the null hypothesis.

The significance level is the probability (**p**) of rejecting the null hypothesis when it is true. Here the p-value is the probability of obtaining results as extreme or more extreme than the observed data if the null hypothesis is true.

Commonly used values for significance level are 0.01, 0.05, or 0.10

If **p-value** $< \alpha$, **reject** the null hypothesis. If **p-value** $\geq \alpha$, **fail to reject (select)** the null hypothesis.

When you wrongly reject the correct null hypothesis, it's called a **type I error**. When you wrongly select an incorrect null hypothesis, it's a **type II error**.

Testing the Alternative Hypothesis

Alternative hypotheses may require a **one-tailed test** or a **two-tailed test (directionality)**. For example,

H1: The **difference in the return** between A and B **> zero**.

To test this alternative hypothesis (H1), we have to test the **mean return**. It should be tested whether the mean return from A is significantly greater or significantly less than that of B i.e. it is tested in both directions. So, it is called a **two-tailed test**.

If H1 is given as

H1: The mean return of A is greater than the mean return of B.

Then testing **in only one direction** is required i.e. it is tested to find out whether the mean return from A is significantly greater than of B. This is called a **one-tailed test**.

Differences between the two types of hypotheses

	Null hypotheses (H_0)	Alternative hypotheses (H_a)
Definition	A claim that there is no effect on the population.	A claim that there is an effect on the population.
Also known as	H_0	H_a H_1
Typical phrases used	. No effect . No difference . No relationship . No change . Does not increase . Does not decrease	. An effect . A difference . A relationship . A change . Increases . Decreases
Symbols used	Equality symbol ($=$, $\geq$, or $\leq$)	Inequality symbol ($\neq$, $<$, or $>$)
$p \leq \alpha$	Rejected	Supported
$p > \alpha$	Failed to reject	Not supported

T-Test

The **t-test** is used to **compare the means** of **two groups** – Parametric test

Take each 25 petals from two different species of iris flowers. Measure their petals' lengths.

H0: The **difference** of mean petal length between these two species is **zero**.

H0 t-test:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{(s^2(\frac{1}{n_1} + \frac{1}{n_2}))}}$$

$$s = \sqrt{((n_1 - 1)s_1^2 + (n_2 - 1)s_2^2)/(n_1 + n_2 - 2)}$$



$\bar{x}_1$ and $\bar{x}_2$ are the means of the two groups being compared

n_1, n_2 : Sample size for group 1 and group 2, respectively.

s_1, s_2 : Standard deviation for group 1 and group 2, respectively.

s is the **joint (pooled)** standard error of the two groups. It is weighted average of s_1 and s_2 . i.e. the group with larger sample size will be given more weightage.

Larger 't' indicates that the difference between two group means are large,

This 't' value is compared with the expected value given in the **standard critical table (t-table)**

If the **'t' value > expected value**, the two species are not same i.e. they are different and hence

H0 is rejected i.e. difference in the mean length exists it is not zero.

Note that the pooled standard deviation should only be used when the standard deviations between the two groups can be assumed to be roughly equal.

Also note that because the pooled standard deviation is a weighted average, it will give more “weight” to the group with the larger sample size. A larger t value shows that the difference between group-means is greater than the pooled standard error, indicating a more significant difference between the groups.

It is also used to determine whether a **process** or treatment actually **has an effect** on the population of interest.

If you want to compare **more than two groups**, or if you want to do multiple pairwise comparisons, use **Analysis of Variance (ANOVA)** test or a **post-hoc** test. You can calculate it manually using a formula or use statistical analysis software.

One-sample, paired, two-sample T-tests:

A population will have many groups or **samples**

- A **one-tailed t-test** is used to know whether one population mean is greater than or less than the other.
- A **two-tailed t-test** is performed to know whether two populations are different from one another.

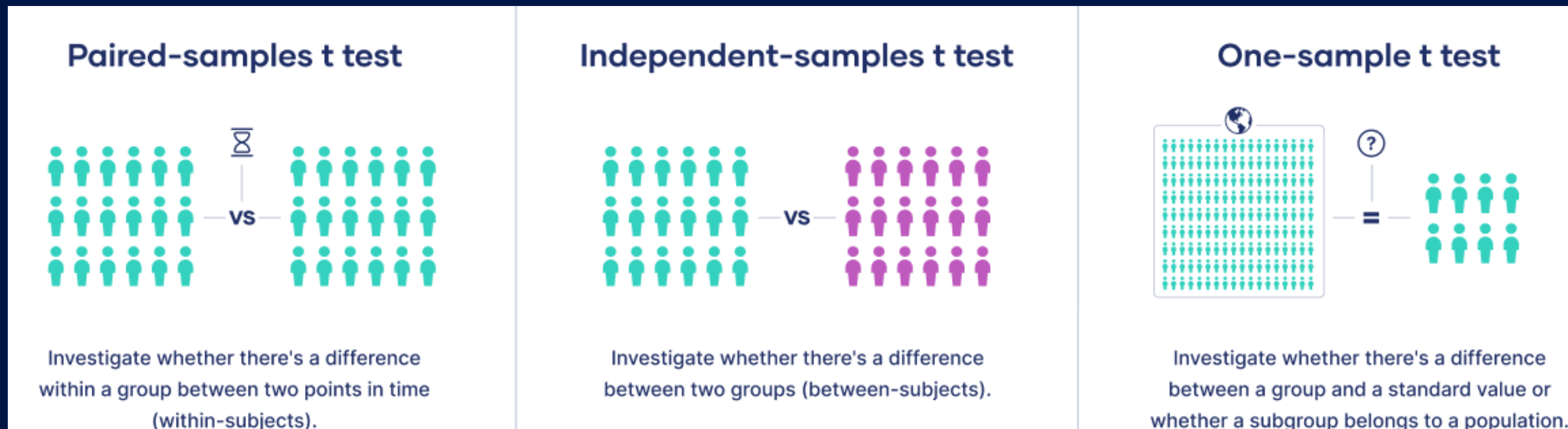
If there is **one group** to be compared against **a standard value** (e.g., comparing the acidity of a liquid to a neutral pH of 7), a **one-sample t-test** may be performed.

If the groups come from **two different populations** (e.g. two different species or people from two different cities), a **two-sample t-test** (independent t-test) is performed. So, this is a **between-subjects** or between-groups design i.e. it compares unrelated measurements taken from separate groups

If the **groups** come from a **single population** (e.g., measuring before and after an experimental treatment), **a paired t-test** is performed. This is a **within-subjects** (repeated measures) **design** i.e. it compares related measures from the **same participants** **between different conditions or times**.

T-Test Contd. ...

In the above example, our interest is to know whether petal length differs by species. The observations come from **two** separate **populations** (separate species), so a **two-sample t-test** is to be performed. If the **direction** of the difference is **not needed** and we want to know whether there is a difference between the two species, then a **one-tailed t-test** may be used.



Critical values of t for two-tailed tests

Significance level (α)

Degrees of freedom (df)	.2	.15	.1	.05	.025	.01	.005	.001
1	3.078	4.165	6.314	12.706	25.452	63.657	127.321	636.619
2	1.886	2.282	2.920	4.303	6.205	9.925	14.089	31.599
3	1.638	1.924	2.353	3.182	4.177	5.841	7.453	12.924
4	1.533	1.778	2.132	2.776	3.495	4.604	5.598	8.610
5	1.476	1.699	2.015	2.571	3.163	4.032	4.773	6.869
6	1.440	1.650	1.943	2.447	2.969	3.707	4.317	5.959
7	1.415	1.617	1.895	2.365	2.841	3.499	4.029	5.408
8	1.397	1.592	1.860	2.306	2.752	3.355	3.833	5.041
9	1.383	1.574	1.833	2.262	2.685	3.250	3.690	4.781
10	1.372	1.559	1.812	2.228	2.634	3.169	3.581	4.587
11	1.363	1.548	1.796	2.201	2.593	3.106	3.497	4.437
12	1.356	1.538	1.782	2.179	2.560	3.055	3.428	4.318

T-Test Contd. ...

Example 1: You want to know whether the mean petal length of iris flowers **differs** according to their species. You find two different species of irises growing in a garden and measure 25 petals of each species.

You can **test the difference** between these two groups using a **null & alternative hypotheses** and **t-test**

- The null hypothesis (H_0) is that the true **difference** between these **group means** is **zero**.
- The alternate hypothesis (H_a) is that the true difference is **different from zero**.

Chi-square Test

It is a **non-parametric** test. So, the data **need not follow normal distribution**.

chi – square
$$X^2 = \sum \frac{(O - E)^2}{E}$$

O is the observed frequency

E is the expected frequency

To decide whether the difference is big enough, compare the chi-square value to a critical value.

It is used to test whether the **observed frequency (O)** distribution of a **categorical variable** is significantly **different (test of independence)** from its **expected frequency (E)** distribution.

Chi-square Test Contd. ...

chi-square goodness of fit test :

Used when there is only **one categorical** variable.

Bird species	Frequency
House sparrow	15
House finch	12
Black-capped chickadee	9
Common grackle	8
European starling	8
Mourning dove	6

Expectations of equal proportions (always)

Null hypothesis (H_0): The bird species visit the bird feeder in **equal proportions**. (i.e. always equal proportions, word equal i.e. **no difference** indicates Null hypothesis)

Alternative hypothesis (H_A): The bird species visit the bird feeder in **different proportions**. (i.e. always different proportions)

Expectations of different proportions (specific period)

Null hypothesis (H_0): The bird species visit the bird feeder in the **same proportions** as the average over the past five years.

Alternative hypothesis (H_A): The bird species visit the bird feeder in **different proportions** from the average over the past five years.

Chi-square Test Contd. ...

chi-square test of independence:

Used when there are **two categorical** variables.

Example: Handedness and nationality – Frequency Distribution Table

Contingency table of the handedness of a sample of Americans and Canadians		
Nationality	Right-handed	Left-handed
American	236	19
Canadian	157	16

If the two variables are independent (unrelated), the probability of one variable for belonging to a certain group is not affected by the other variable.

Null hypothesis (H_0): The proportion of left-handed people **is the same** for Americans and Canadians.

Alternative hypothesis (H_A): The proportion of left-handed people **differs between nationalities**.

Several other types of chi-square tests are not Pearson's chi-square tests, including the test of a single variance and the likelihood ratio chi-square test.

Chi-square Test Contd. ...

Two-way table

Frequency Row Pct	Product Category					Total
	1	2	3	4	5	
Small	1 11.11	2 22.22	2 22.22	3 33.33	1 11.11	9
Medium	1 1.43	5 7.14	22 31.43	30 42.86	12 17.14	70
Large	0 0.00	0 0.00	0 0.00	2 100.00	0 0.00	2
Total	2	7	24	35	13	81

Frequency Missing = 77

Chi-Square test:

$$\chi^2 = \sum (O - E)^2 / E$$

$$E = \frac{\text{row total} \times \text{column total}}{\text{sample size}}$$

Product 1 has 1 small category 1 medium category and there is no large category. Similarly, product 2 has 2 small categories and so on . . .

The expected count (E) for product **category 1 to be of small** size = $9 \times 2 / 81 = 2/9 = 0.22$

It is derived by taking the row total for Size (9) times the column total for Product category (2) and then dividing by the sample size (81). This procedure is conducted for each cell.

Correlation

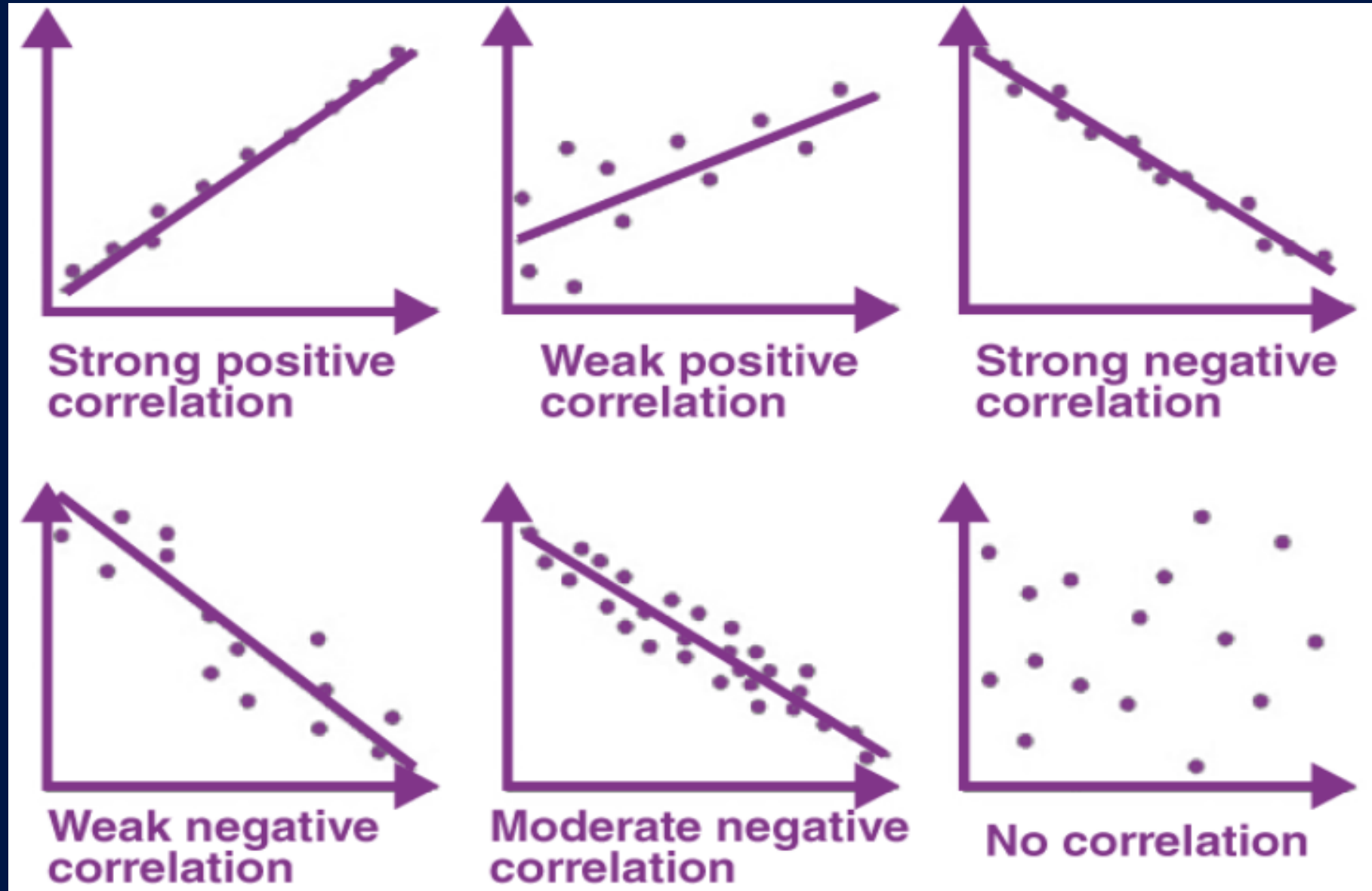
Correlation means there is a **statistical association** between variables. These variables change together i.e. they **covary**.

However, this covariation is not necessarily due to a direct or indirect causal link.

A correlation coefficient is a number between -1 and 1 that tells you the strength and direction of how similar the variables are.

Correlation coefficient value	Correlation type	Meaning
1	Perfect positive correlation	When one variable changes, the other variables change in the same direction.
0	Zero correlation or No correlation	There is no relationship between the variables.
-1	Perfect negative correlation	When one variable changes, the other variables change in the opposite direction.

Correlation Contd. ...



Pearson Correlation Coefficient between two variables x and y

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

Where n = Quantity of Information

$\sum x$ = Total of the First Variable Value

$\sum y$ = Total of the Second Variable Value

$\sum xy$ = Sum of the Product of first & Second Value

$\sum x^2$ = Sum of the Squares of the First Value

$\sum y^2$ = Sum of the Squares of the Second Value

Positive Correlation ($r > 0$) – when the values of the two variables move in the same direction so that an increase/decrease in the value of one variable is followed by an increase/decrease in the value of the other variable.

Negative Correlation ($r < 0$) – when the values of the two variables move in the opposite direction so that an increase/decrease in the value of one variable is followed by a decrease/increase in the value of the other variable.

No Correlation ($r = 0$) – when there is no linear dependence or no relation between the two variables.

Types of Predictive Models

A predictive analytics model is essentially a set of algorithms that discovers patterns in data and uses those patterns to predict beneficial outcomes

1. Classification Model

These models perform categorical analysis on data and predicts the class for new data.

These models can be retrained with new data to obtain useful and detailed insights.

Classification models are customizable and are helpful across industries, including banking and retail.

2. Clustering Model

The clustering model divides the data into groups based on common characteristics.

Hard clustering determines if each data point belongs to a cluster and soft clustering allocates a probability to each data point.

A clustering model can help businesses plan marketing campaigns for certain groups of customers.

3. Forecast Model

It provides new data values based on historical data insights.

Forecast models also generate numerical values in historical data if none are present.

It can manage multiple parameters at a time.

A call center can use forecast analytics to predict how many support calls they will receive in a day, or a retail store can forecast inventory for the upcoming holiday sales periods, etc.

4. Outliers Model

The outlier model detects or deals with **anomalous data** items within a dataset.

Outlier models are essential in industries like retail and **finance**, where **detecting abnormalities** can save **businesses** millions of dollars.

Outlier models can quickly identify anomalies, so they are efficient **in fraud detection**.

5. Time Series Model

These models analyze datasets where the input parameter is **time sequences**.

The time series model develops a numerical value that **predicts the trends within a specific period** by combining multiple data points (from the previous year's data).

A Time Series model outperforms traditional ways of calculating a variable's progress because it may forecast for numerous regions or projects at once or focus on a single area or task, depending on the organization's needs.

Helpful if organizations need to **know how a specific variable changes over time**.

For example, if a small business owner wishes to track sales over the last four quarters, they will need to use a Time Series model. It can also look at external factors like seasons or periodical variations that could influence future trends.

Predictive Models Challenges Faced:

One of the most common problems with predictive models is that they **require data** preparation with **high accuracy and thoroughness**.

Achieving high accuracy in data preparation can be difficult, especially when you have to consider **multiple algorithms** and **their differences in data preparation** requirements and assumptions.

Each predictive algorithm has **unique data preparation** requirements and assumptions, which must be known and addressed before building the model.

Another problem is that **knowing which algorithm is best** in advance is hard. Even for experienced practitioners, it's impossible to tell whether an algorithm will be effective without trial and error.

The biggest problem with using predictive **analytics models** is that **failure is common**. It's not always clear why models fail, but it can be frustrating when you know you have a great model and it doesn't work.

It can also be **hard** to tell **how good a model** is in advance; you have to run it on **actual data** to know if it will work well, which means waiting until after something has happened before you know if your model performed correctly.

Univariate Analysis

If each observation or **data point** corresponds to a **single variable**, then it is called **univariate data**. Consider the heights of seven students recorded in the table given below, there is only one **variable**, which is **height**.

Heights (in cm)	164	167.3	170	174.2	178	180	186
-----------------	-----	-------	-----	-------	-----	-----	-----

No Relationships: Univariate analysis focuses solely on **describing and summarizing** the distribution of the **single variable**. It does not explore **relationships between variables** or attempt to identify causes.

Descriptive Statistics: Descriptive statistics, such as measures of **central tendency** (mean, median, mode) and **measures of dispersion** (range, standard deviation), are commonly used in the analysis of univariate data.

Visualization: Histograms, **box plots**, and other **graphical representations** are often used to visually represent the distribution of a single variable.

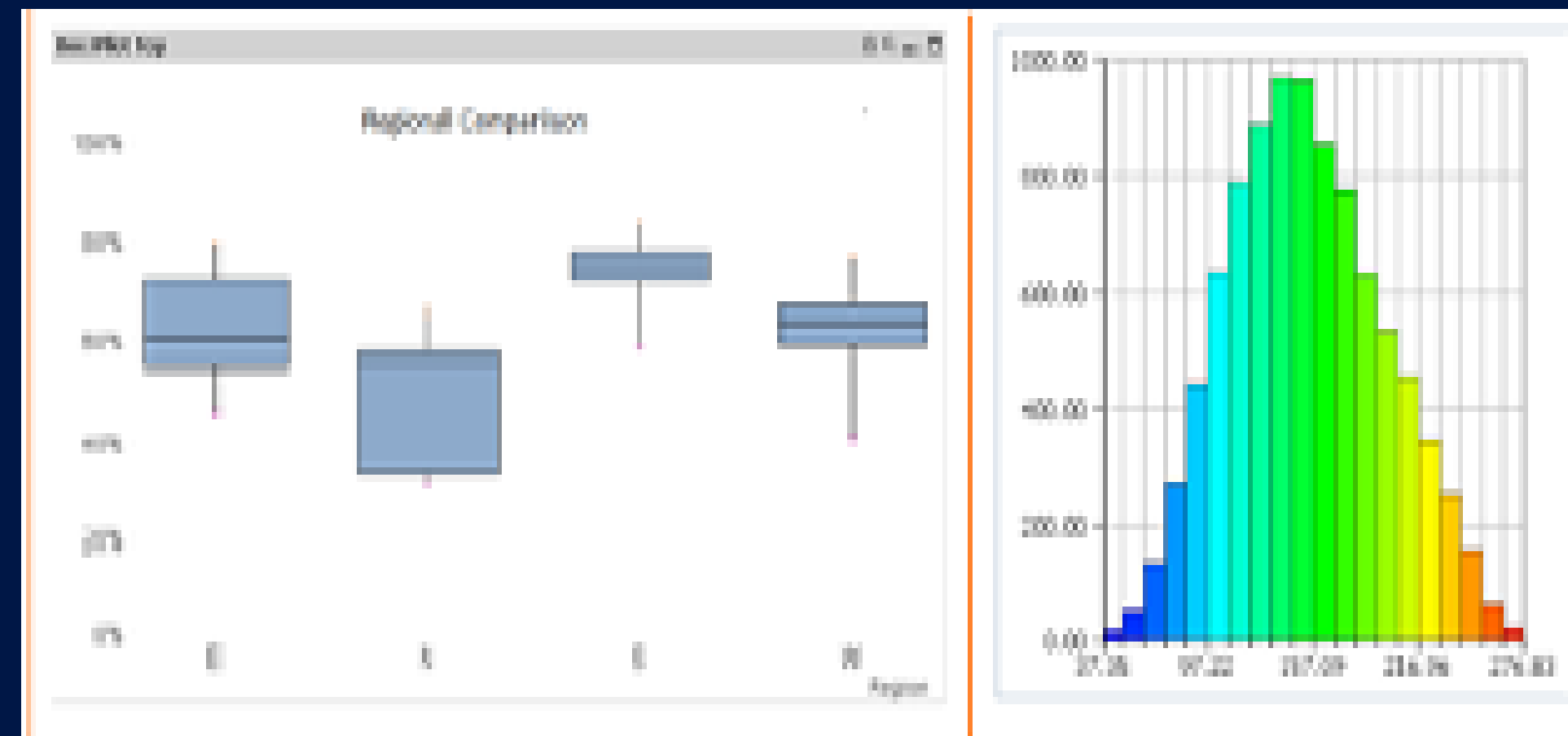
Univariate Analysis Contd. ...

Performing univariate analysis will depend on whether the variable type is categorical or continuous.

Continuous Variables:- In the case of continuous variables, we need to understand the **central tendency** and **spread** of the variable.

Categorical Variables: use a **frequency table** to understand the distribution of each category. We can also read as a percentage of values under each category. It can be measured using two metrics, **Count** and **Count%** against each category. **Bar chart** can be used to visualize them.

Central Tendency	Measure of Dispersion	Visualization Methods
Mean	Range	Histogram
Median	Quartile	Box Plot
Mode	IQR	
Min	Variance	
Max	Standard Deviation	
	Skewness and Kurtosis	



Bivariate Analysis

Bivariate data involves **two different variables**, and the analysis of this type of data focuses on understanding the **relationship** or association **between these two variables**.

An example of bivariate data can be **temperature** and **ice cream sales** in the summer season.

Suppose the temperature and ice cream sales are the two variables of the bivariate data given in the table below. Here, the relationship is visible from the table that temperature and sales are **directly proportional** to each other and thus related because as the temperature increases, the sales also increase.

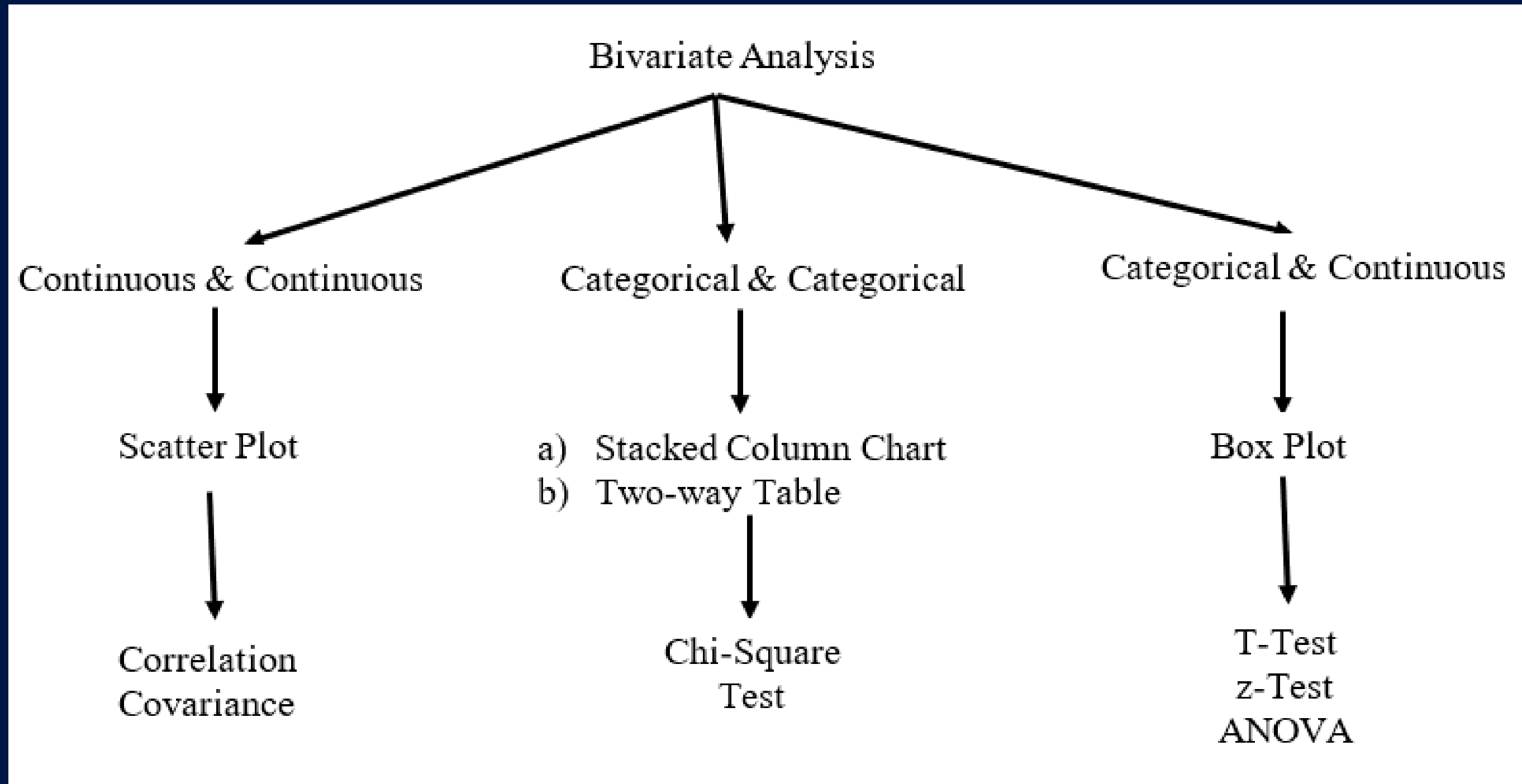
Relationship Analysis: The primary goal of analyzing bivariate data is to understand the **relationship between the two variables**. This relationship could be **positive** (both variables increase together), **negative** (one variable increases while the other decreases), or show **no clear pattern**.

Temperature	Ice Cream Sales
20	2000
25	2500
35	5000

Correlation Coefficient: A **quantitative measure** called the correlation coefficient is often used to quantify the strength and direction of the linear relationship between two variables. The correlation coefficient ranges from -1 to 1.

Bivariate Analysis Contd. ...

Scatter plots: A common visualization tool for bivariate data is a **scatterplot**, where each data point represents a pair of values for the two variables. Scatterplots help visualize patterns and trends in the data.



Bivariate Analysis Contd. ...

i) Continuous & Continuous: when two continuous variables are analyzed, the **pattern** of the scatter plot indicates the relationship between variables. The relationship can be linear or non-linear.

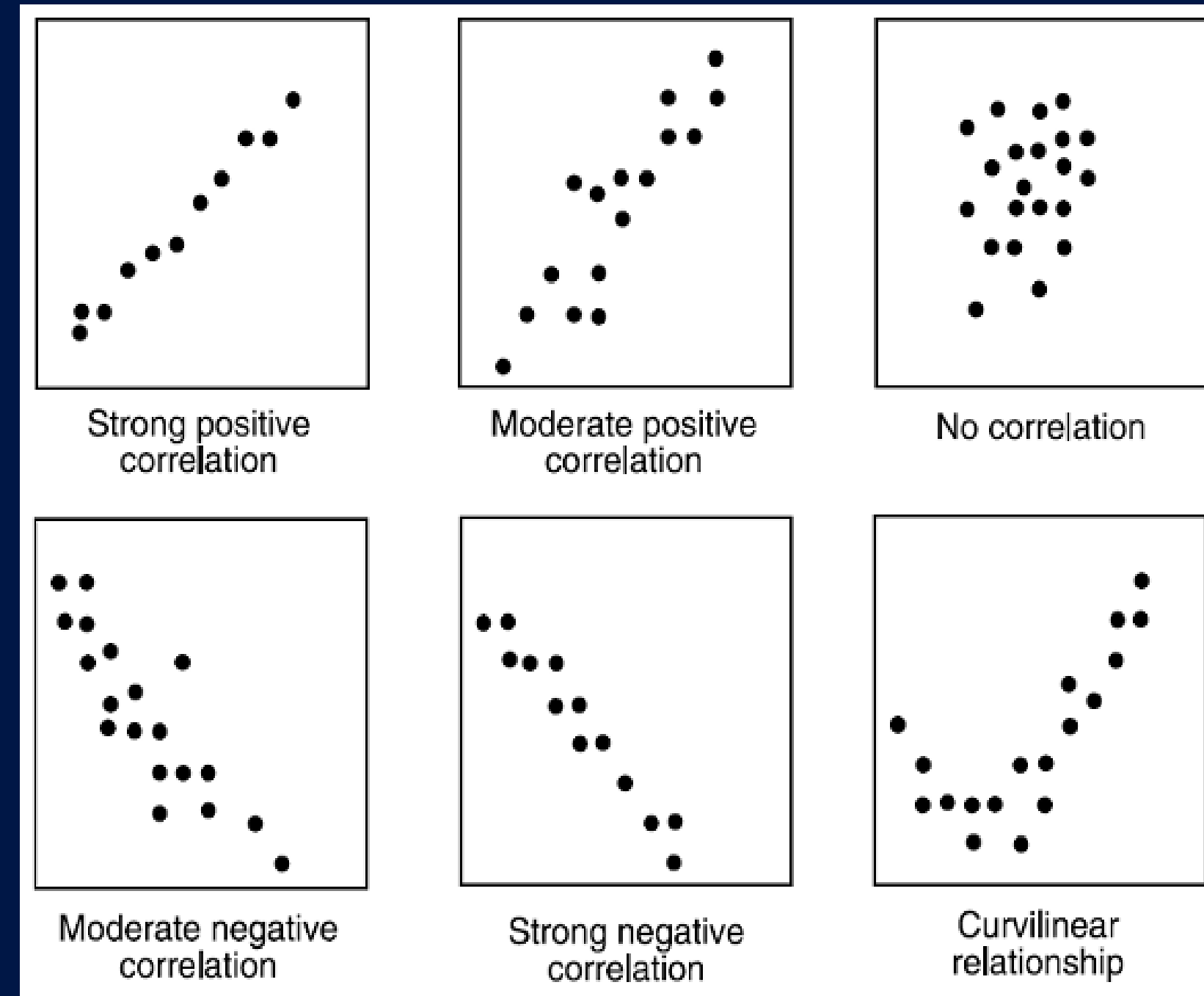
The scatter plot shows the relationship between two variables but **does not indicate the strength** of the relationship between them.

To find the strength of the relationship, we use Correlation. Correlation varies between -1 and +1.

-1: perfect negative linear correlation

+1: perfect positive linear correlation and

0: No correlation



Bivariate Analysis Contd. ...

Correlation = Covariance(X,Y) / SQRT(Var(X)* Var(Y))

Population Covariance Formula

$$\text{Cov}(x,y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N}$$

Sample Covariance

$$\text{Cov}(x,y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N - 1}$$

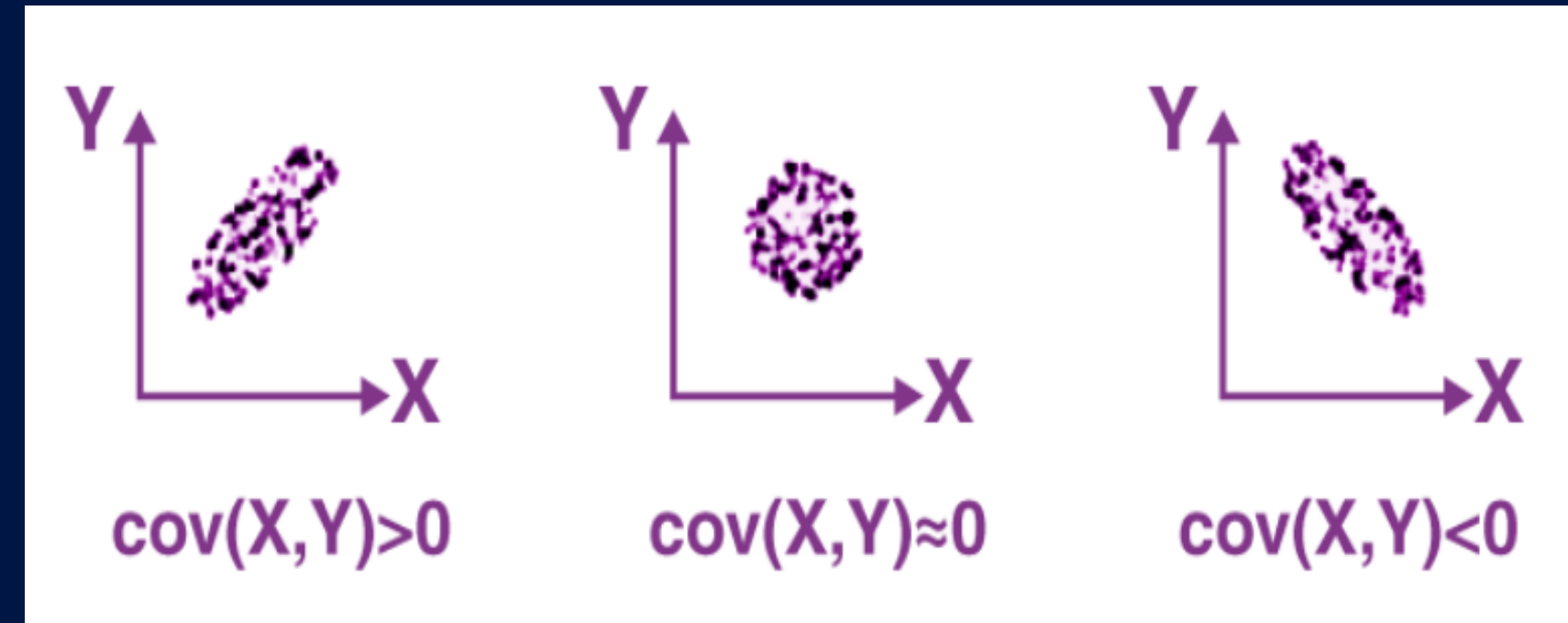
x_i = data value of x

y_i = data value of y

$\bar{x}$ = mean of x

$\bar{y}$ = mean of y

N=number of data values.



ii) Categorical & Categorical: Chi-Square Test is used to derive the statistical significance of the relationship between the variables.

We can use the following two methods to estimate and visualize:

a) Two-way table: We can start analyzing the relationship by creating a two-way table of count and count%. The **rows** represent the category of **one variable** and the **columns** represent the categories of the **other variable**. We show the count or count% of observations available in each combination of row and column categories.

Bivariate Analysis Contd. ...

a) Two-way table

Frequency Row Pct	Product Category					Total
	1	2	3	4	5	
Small	1 11.11	2 22.22	2 22.22	3 33.33	1 11.11	9
Medium	1 1.43	5 7.14	22 31.43	30 42.86	12 17.14	70
Large	0 0.00	0 0.00	0 0.00	2 100.00	0 0.00	2
Total	2	7	24	35	13	81

Frequency Missing = 77

Chi-Square test:

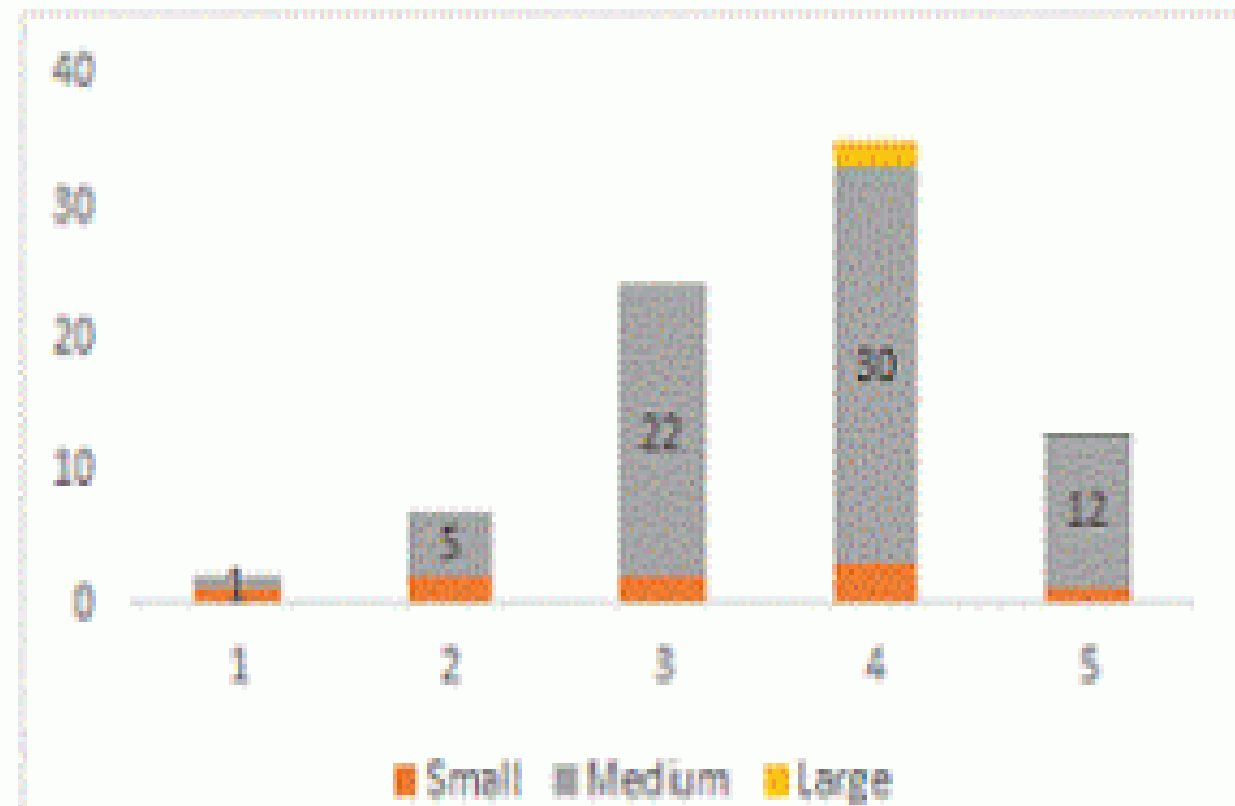
$$\chi^2 = \sum (O - E)^2 / E$$

$$E = \frac{\text{row total} \times \text{column total}}{\text{sample size}}$$

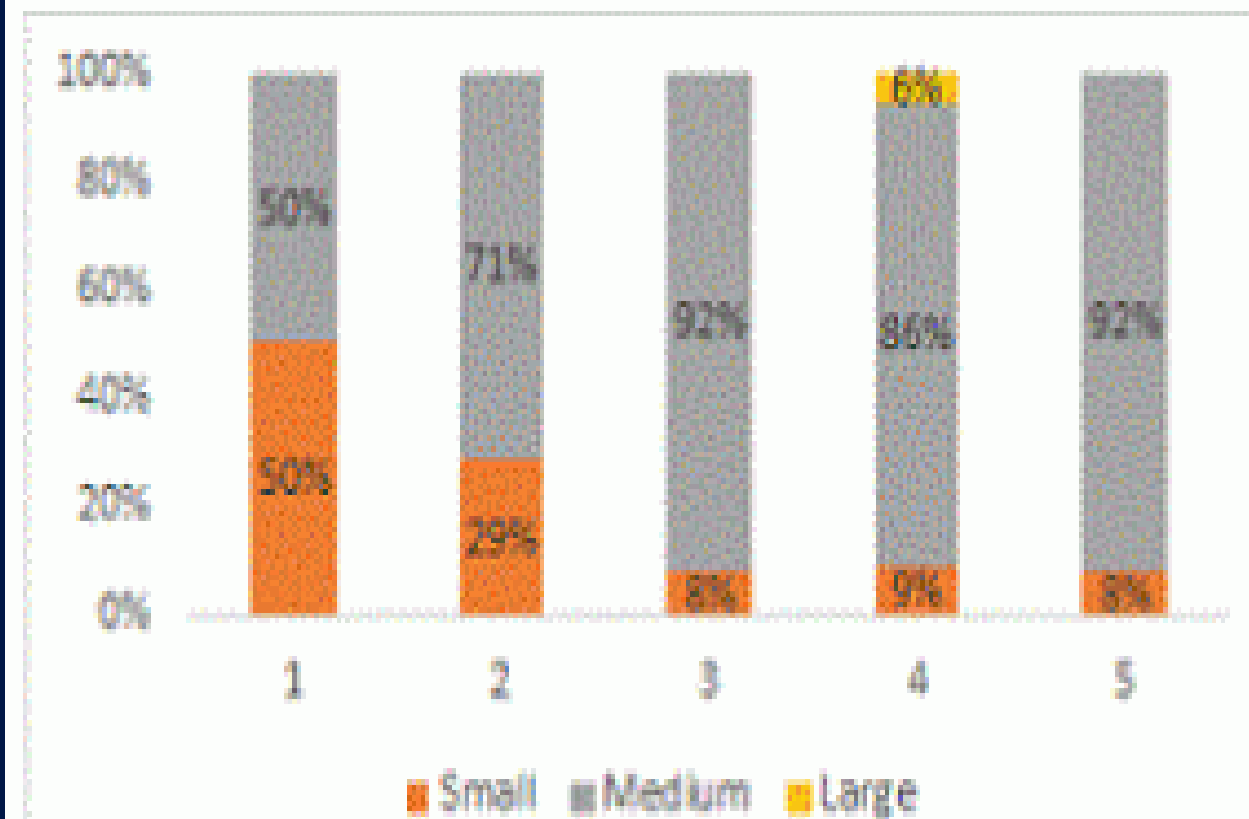
The expected count (E) for product category 1 to be of small size is 0.22. It is derived by taking the row total for Size (9) times the column total for Product category (2) and then dividing by the sample size (81). This procedure is conducted for each cell.

Bivariate Analysis Contd. ...

b) Stacked Column Chart: This method is more of a visual form of a Two-way table.



Column Value Count



Column Values in Percentage

Frequency Row Pct	Product Category					Total
	1	2	3	4	5	
Small	1 11.11	2 22.22	2 22.22	3 33.33	1 11.11	9
Medium	1 1.43	5 7.14	22 31.43	30 42.86	12 17.14	70
Large	0 0.00	0 0.00	0 0.00	2 100.00	0 0.00	2
Total	2	7	24	35	13	81

Frequency Missing = 77

Bivariate Analysis Contd. ...

iii) **Categorical & Continuous:** We can draw box plots for each level of categorical variables. We can use T-test, z-test or ANOVA

These tests assess whether the **mean of two groups** is statistically different from each other or not.

$$Z = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

where:

- $\bar{X}_1, \bar{X}_2$: Averages
- S_1^2, S_2^2 : Variances
- N_1, N_2 : Counts
- t : has t distribution with $N_1 + N_2 - 2$ degree of freedom

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{S^2 \left(\frac{1}{N_1} + \frac{1}{N_2} \right)}}$$

$$S^2 = \frac{(N_1 - 1)S_1^2 + (N_2 - 1)S_2^2}{N_1 + N_2 - 2}$$

Missing Values

Reasons for the Occurrence of Missing Values:

Data Extraction: There may be problems with the extraction process. In such cases, we should double-check for correct data with **data guardians**. Some **hashing procedures** can also be used to make sure data extraction is correct. Errors at the data extraction stage are typically easy to find and can be corrected easily as well.

Data collection: These errors occur at the time of data collection and are harder to correct. They can be categorized into three types:

a)Missing Completely At Random (MCAR) : The value could be missing due to **human error**, some system/equipment failure, loss of sample, or some unsatisfactory technicalities while recording the values.

In this case, there is **no relationship between the missing data and any other values** observed or unobserved (the data that is not recorded) within the given dataset. That is, missing values are completely independent of other data. There is no pattern.

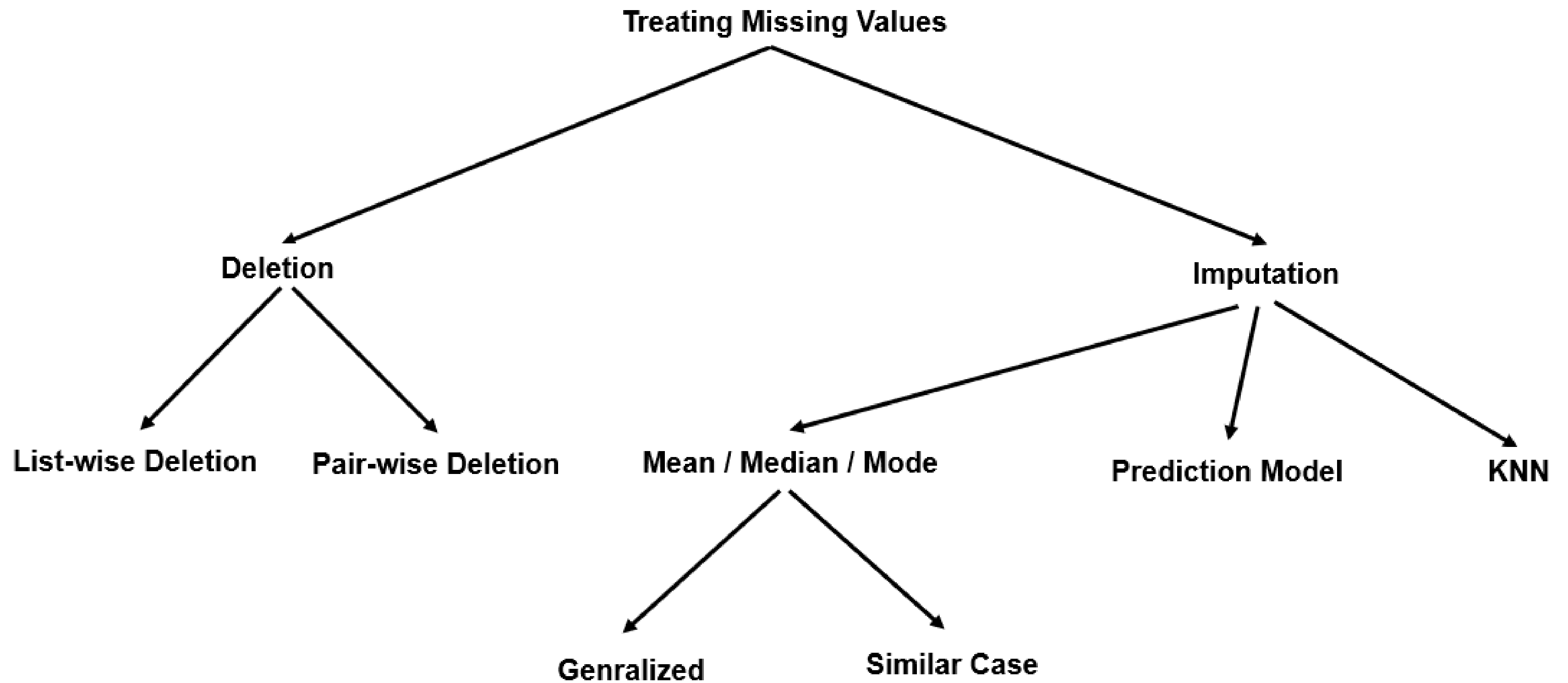
For Example, Suppose in a library there are some overdue books missing. The reason might be a **human error** like the librarian forgetting to type in. It should not be assumed as it's a rare case.

b) Missing at random (MAR) : There is **some relationship between the missing data** and other values/data and there is some pattern in the missing values.

For example, if you check the survey data, you may find that all the people have answered their 'Gender,' but '**Age**' values are mostly missing for people who have answered their 'Gender' as 'Female.' (The reason being most of the **females** don't want to reveal their age.). So, the probability of data being missing depends only on the **observed value** or data. In this case, the variables 'Gender' and 'Age' are related, **but you can not predict the missing value** itself. In this case, the statistical analysis might result in bias. Getting an unbiased estimate of the parameters can be done only **by modeling the missing data**.

c) Missing not at Random (MNAR) (or) Missing that depends on unobserved data: Data will be missing **because they are not observed**. In a medical study, if a particular diagnostic causes discomfort, then there is a higher chance of dropping out of the study. This missing value is not random unless we have included "**discomfort**" as an input variable for all patients.

Treating the Missing Values



List wise deletion		
Gender	Manpower	Sales
M	25	343
F	.	280
M	33	332
M	.	272
F	25	.
M	29	326
.	26	259
M	32	297

Pair wise deletion		
Gender	Manpower	Sales
M	25	343
F	.	280
M	33	332
M	.	272
F	25	.
M	29	326
.	26	259
M	32	297

i) Deletion: Deletion methods are used when the nature of missing data is “Missing completely at random” else non-random missing values can bias the model output. There are two types of deletion methods.

List-wise deletion: **Delete the observations if any value is missing.** Simplicity is the major advantage of this method, but this method reduces the power of the model because it reduces the sample size.

Pair-wise deletion: **we perform analysis with all cases** in which the variables of interest are present. Advantage of this method is that it keeps as many cases available for analysis. The disadvantage of this method is that it uses **different sample sizes for different variables.**

Imputation is a method to fill in the missing values with estimated ones. The objective is to employ known relationships that can be identified from the valid values of the data set to assist in estimating the missing values.

ii) Mean/ Mode/ Median Imputation

It replaces the missing data for a given attribute with the mean or median (quantitative attribute) or mode (qualitative attribute) of all known values of that variable (column). Mean / Mode / Median imputation is one of the most frequently used methods. It can be of two types:-

a) **Generalized Imputation:** In this case, we calculate the **mean or median for all non-missing values** of that variable and then replace the missing value with the mean or median. As in the above table, the values for the variable “Manpower” are missing so we take the average (mean) of all non-missing values of “**Manpower**” ($(25+33+25+29+26+32)/6 = 28.33$) and then replace the missing values with it.

b) **Similar case Imputation:** In this case, we calculate the average for gender “**Male**” (29.75) and “**Female**” (25) individually of non-missing values then replace the missing value based on gender. For “Male”, we will replace the missing values of manpower with 29.75 and for “Female” with 25.

iii) Using a Prediction Model: The prediction model is one of the sophisticated methods for handling missing data. Here, we create a predictive model to estimate values that will substitute the missing data. In this case, we divide our data set into two sets. One **set with no missing values** for the variable and another one **with missing values**. The first data set is used for **training** the model. The second data set with missing values is used as a **test data** set. The **variable with missing values** is treated as the **target variable**. Next, a model is created to predict the target variable based on other attributes of the training data set and populate missing values of the test data set. We can use **regression, ANOVA, Logistic regression and various modeling techniques to perform this**.

There are 2 drawbacks to this approach:

1. The values estimated or predicted by the model are usually **more well-behaved** than the **true values**
2. If there are no relationships between **attributes (variables) in the data set** and the **attribute with missing values**, then the model will not be precise for estimating missing values.

iv) K-Nearest Neighbour (KNN) Imputation: The missing value of an attribute is imputed from a similar group. The similarity of the two attributes is determined using a distance function.

a. Advantages:

- i. k-nearest neighbour can predict both qualitative & quantitative attributes.
- ii. The creation of a predictive model for each attribute with missing data is not required
- iii. Attributes with multiple missing values can be easily treated
- iv. The correlation structure of the data is taken into consideration

b. Disadvantage:

- i. KNN algorithm is very time-consuming in analyzing large databases. It searches through all the datasets looking for the most similar instances.
- ii. The choice of k-value is very critical. A higher value of k would include attributes that are significantly different from what we need whereas a lower value of k implies missing out on significant attributes.

Importance of Treating Missing Values:

Missing data in the training data set can reduce the power/fit of a model or can lead to a biased model

Missing data is defined as the values or data that is not stored (or not present) for some variable/s in the given dataset.

Name	Weight	Gender	Play Cricket/ Not
Mr. Amit	58	M	Y
Mr. Anil	61	M	Y
Miss Swati	58	F	N
Miss Richa	55		Y
Mr. Steve	55	M	N
Miss Reena	64	F	Y
Miss Rashmi	57		Y
Mr. Kunal	57	M	N

Gender	#Students	#Play Cricket	%Play Cricket
F	2	1	50%
M	4	2	50%
Missing	2	2	100%

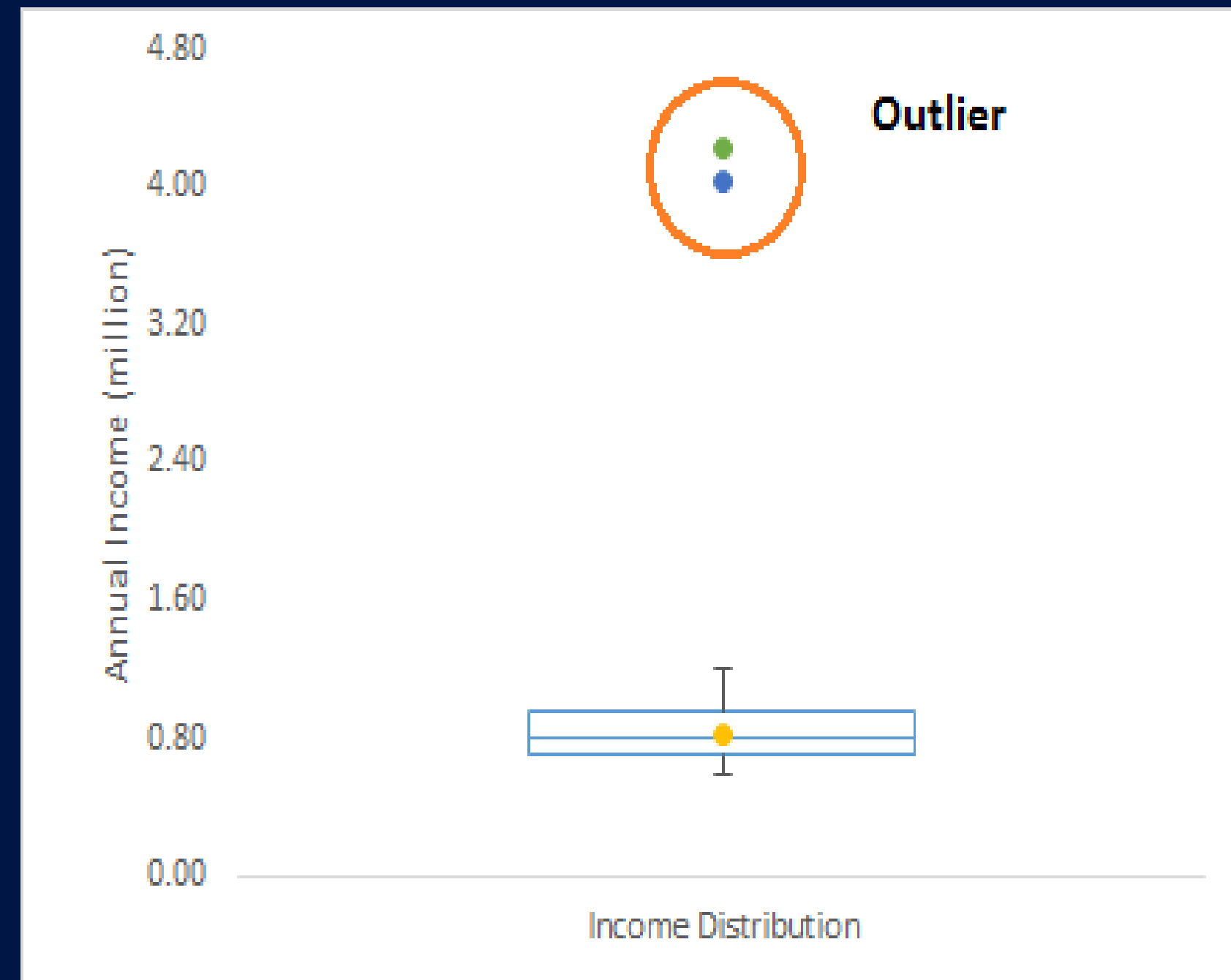
In the first table, we have not treated missing values i.e. we are not considering the rows that have missing values.

Wrong Inference for Women Playing Cricket

OUTLIERS

An outlier is an observation that appears far away and diverges from an overall pattern in a sample. It needs close attention else it can result in wildly wrong estimations. Outliers tend to make the data skewed and reduce accuracy. For example, we do customer profiling and find out that the average annual income of customers is \$0.8 million. But, two customers are having annual incomes of \$4 and \$4.2 million. These two customers' annual income is much higher than the rest of the population. These two observations will be seen as Outliers.

In the box plot shown in the figure, the bottom-most line in the box plot indicates that the data that falls below this line are at a lower 25% i.e. the quartile 1 (Q1) data. Similarly, the data between the second and first lines fall between 25% and 50% i.e. 2nd quartile (Q2) data. The data that falls between the 3rd line and the 4th line are the Quartile 3(Q3) data. The data that fall above the 4th line are the Q4 data. The data that falls above 1.5 times the minimum value of Q4 are considered outliers. Similarly, the data that falls below 1.5 times the maximum value of Q1 are considered outliers.



Detecting the Outliers:

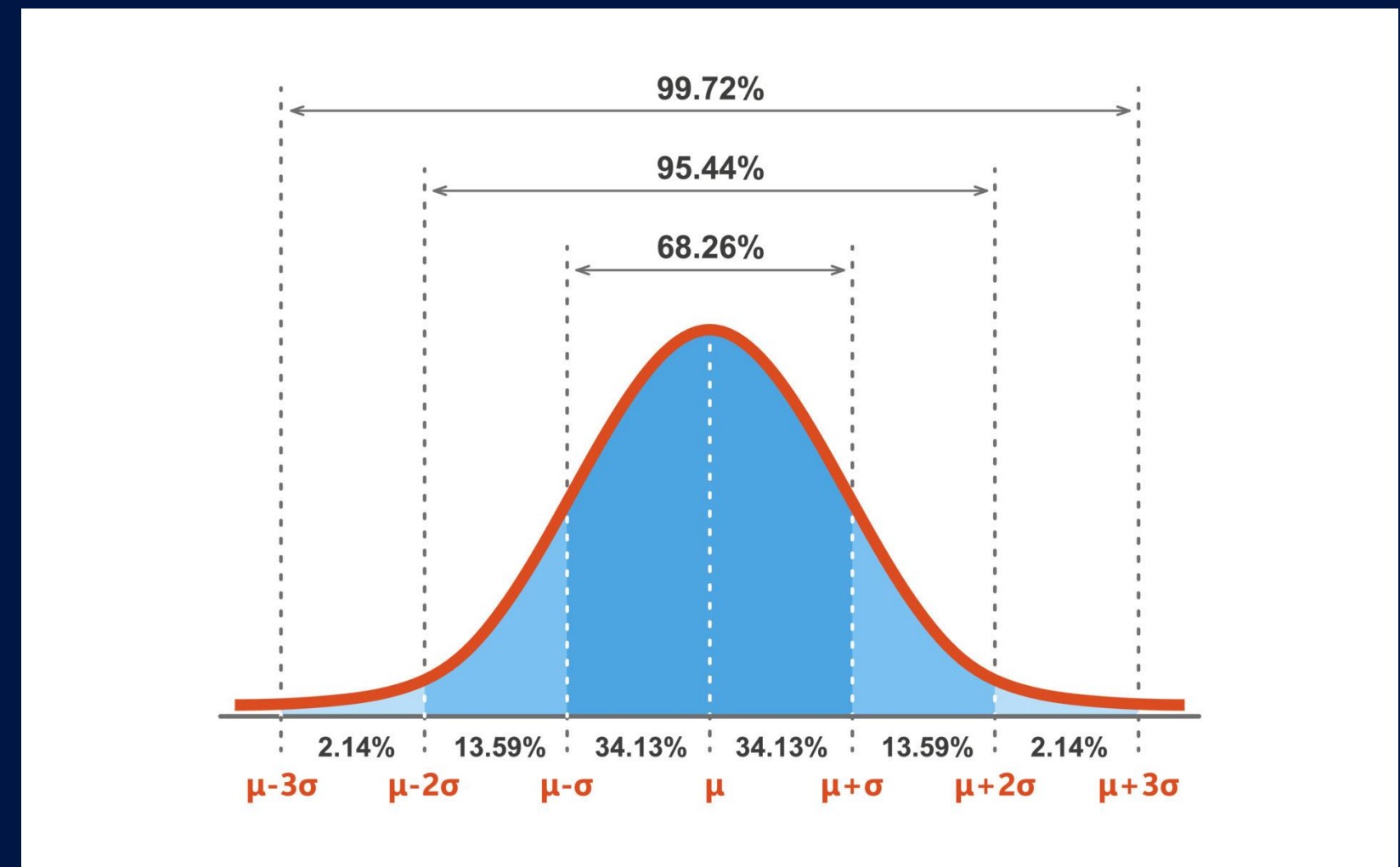
Outlier detection is merely a special case of the examination of data for influential data points and it also depends on the business's understanding. **Capping methods** are techniques used to set upper and lower bounds (caps) on data values to remove outliers.

1. The most commonly used method to detect outliers is **visualization**. We use various visualization methods, like **Box-plot**, **Histogram**, and **Scatter Plot**. Some analysts also use various thumb rules to detect outliers.
2. Detecting outliers in data science using **Box-plot** involves identifying data points that deviate significantly from the majority of the data. This method is using the **Interquartile Range (IQR)**.

Steps to Identify Outliers using Box Plot

- i) $IQR = Q3 - Q1$ Q3- 75th percentile and Q1 - 25th percentile of the data
- ii) Lower Bound: $Q1 - 1.5 * IQR$
Upper Bound: $Q3 + 1.5 * IQR$
- iii) Outlier: Any data point below the lower bound or above the upper bound is considered an outlier.

3. The third approach is based on the assumption that data follows a **normal distribution**. A normal distribution, also known as a **Gaussian distribution or bell curve**, is a continuous probability distribution that is symmetric around its mean, which is also its median and mode. In a normal distribution, approximately 68% of the data falls within one standard deviation of the mean, about 95% falls within two standard deviations, and approximately 99.7% falls within three standard deviations. Therefore, data points beyond this 3σ range are considered unusual or extreme.



Steps to Identify Outliers using mean & Standard deviation

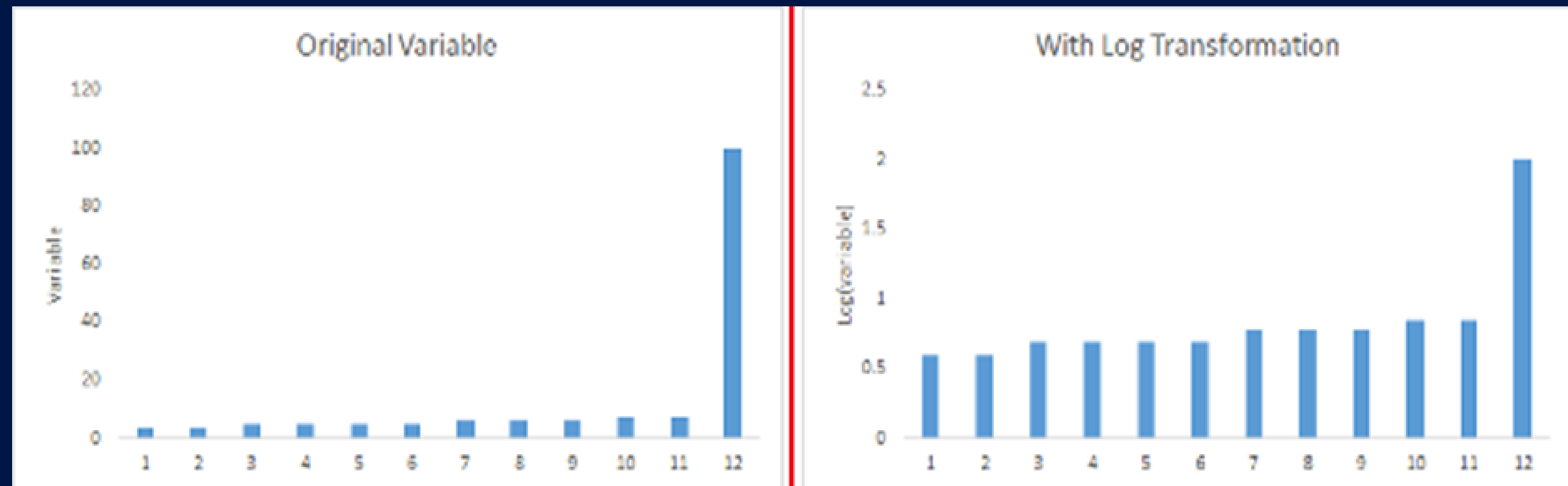
- i) Calculate the Mean (μ) and Standard Deviation (σ) of the data. The mean indicates the average and the standard deviation **measures the spread of the data**.
- ii) Any data point that is more than three standard deviations away from the mean is considered an outlier.
 - a) Lower bound = $\mu - 3\sigma$
 - b) Upper bound = $\mu + 3\sigma$
- iii) Identify Outlier: Any point below the lower bound or above the upper bound

4. **Percentile Cap Method:** Any value out of the range of 5th and 95th percentile can be considered as an outlier

Treating the Outliers:

Deleting observations: We delete outlier values if it is due to data entry error, data processing error or outlier observations are very small in numbers. We can also use trimming at both ends to remove outliers.

Transforming and binning values: Transforming variables can also eliminate outliers. Natural **log of a value** reduces the variation caused by extreme values. **Binning** is also a form of variable transformation. **Decision Tree algorithm** allows to deal with outliers well due to binning of variable. We can also use the process of assigning weights to different observations.



Imputing: Like the imputation of missing values, we can also impute outliers. We can use **mean, median, and mode** imputation methods. Before imputing values, we should analyse if it is **natural outlier or artificial**. If it is **artificial**, we can go with **imputing values**. We can also use **statistical model to predict values** of outlier observation and after that we can impute it with predicted values.

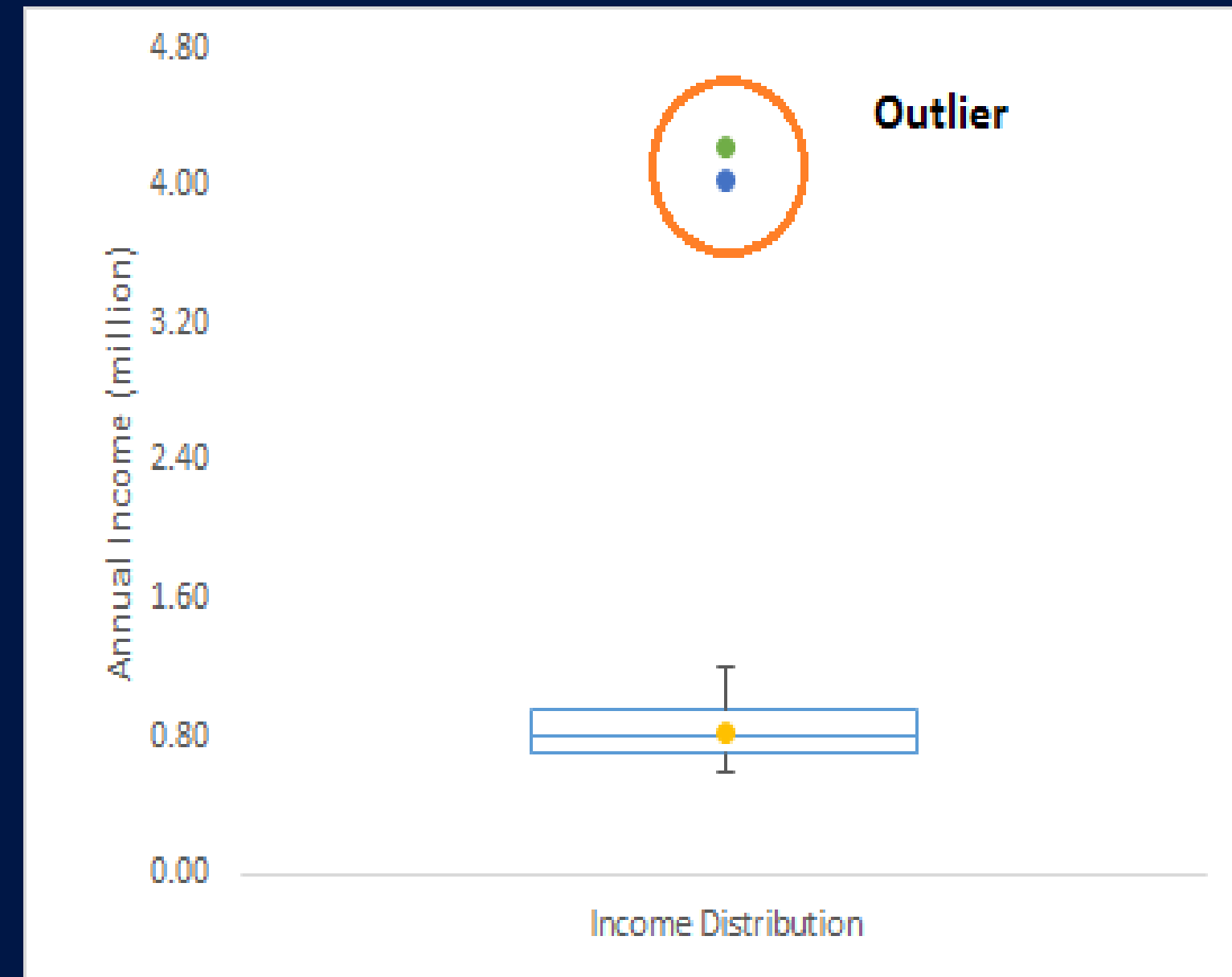
Treat separately: If there are **significant number of outliers**, we should treat them **separately** in the statistical model. One of the approaches is to treat both groups as two **different groups** and build individual models for both groups and then combine the output.

OUTLIERS - TYPES

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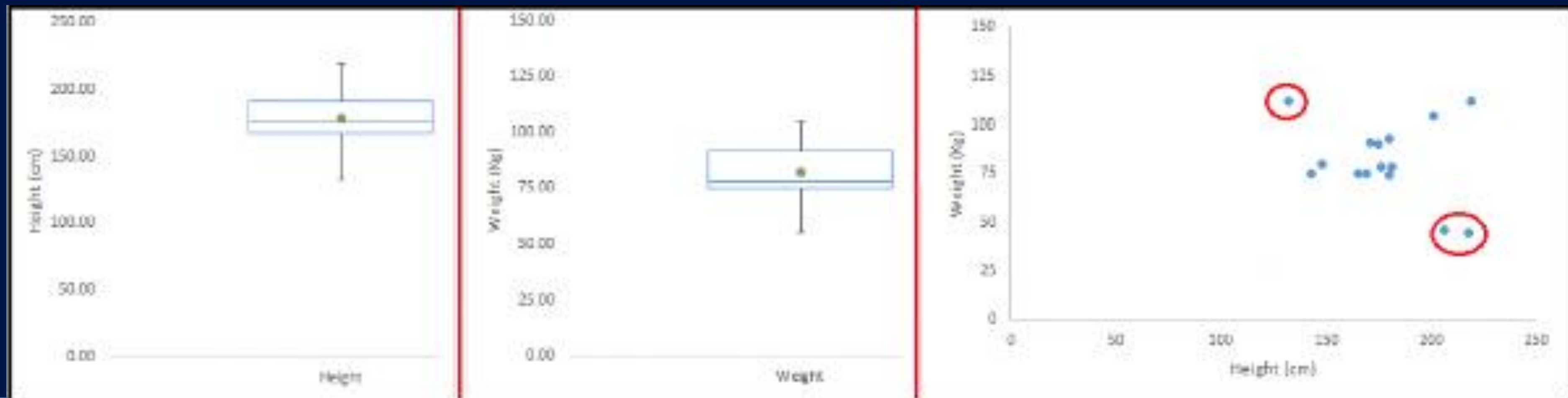


Outliers can be of two types: Univariate and Multivariate.

The example discussed is a **univariate outlier**. These outliers can be found when we look at the distribution of a single variable i.e. along one dimension.

Multi-variate outliers are outliers in an n-dimensional space.

For example, let us say we are understanding the relationship between height and weight. Below, we have univariate and bivariate distributions for Height and weight. Take a look at the box plot. We do not have any outliers. Now look at the scatter plot (3rd plot). Here, we have two values below the average and one above the average in a specific segment of weight and height.



Reasons or Causes of Outliers

The method to deal with outliers would then depend on the **reason (Causes)** for their occurrence. Causes of outliers can be classified into two broad categories:

1. Artificial (Error) / Non-natural

2. Natural.

1. Artificial (Error) / Non-natural

a) **Data Entry Errors:-** Human errors such as errors caused during data collection, recording, or entry can cause outliers in data. For example, the Annual income of a customer is **\$100,000**. Accidentally, the data entry operator puts an **additional zero** in the figure. Now the income becomes **\$1,000,000** which is 10 times higher. this will be the **outlier value** when compared with the rest of the population.

b) **Measurement Error:** It is the most common source of outliers. This is caused when the measurement instrument used turns out to be faulty. For example: There are **10 weighing machines**. **9 of them are correct**, 1 is faulty. The weight measured by people on the **faulty machine will be higher / lower** than the rest of the people in the group. The weights measured on faulty machines can lead to outliers.

Reasons or Causes of Outliers

c) Experimental Error: Another cause of outliers is experimental error. For example: In a 100m sprint of 7 runners, one **runner missed out** on concentrating on the **'Go' call** which caused him to start late. Hence, this caused the runner's run time to be more than other runners. His total run time can be an outlier.

d) Intentional Outlier: This is commonly found in self-reported measures that involve sensitive data. For example: **Teens** would typically under report the **amount of alcohol** that they consume. Only a fraction of them would report actual value. Here **actual values might look like outliers** because rest of the teens are under reporting the consumption.

e) Data Processing Error: Whenever we perform data mining, we extract data from **multiple sources**. Some **manipulation or extraction errors** may lead to outliers in the dataset.

f) Sampling Error: Let we have to measure the **height of athletes**. By mistake, we include a few **basketball players** in the sample. This inclusion is likely to cause outliers in the dataset.

2. Natural Outlier: When an outlier is not **artificial (due to error)**, it is a natural outlier. For example noticing that the performance of top **50 financial advisors was far higher than rest** of the population. It was not due to an error. Hence, whenever we perform any data mining activity with advisors, we have to treat this segment separately. They should not be omitted.

Impact of Outliers on a Dataset:

Outliers can drastically change the results of the data analysis and statistical modeling. There are numerous unfavourable impacts of outliers in the data set:

- ❖ It increases the error variance and reduces the power of statistical tests
- ❖ If the outliers are non-randomly distributed, they can decrease normality (normal distribution, also known as a Gaussian distribution or bell curve)
- ❖ They can bias or influence estimates that may be of substantive interest
- ❖ They can also impact the basic assumption of Regression, ANOVA and other statistical model assumptions.

Without Outlier	With Outlier
4, 4, 5, 5, 5, 5, 6, 6, 6, 7, 7	4, 4, 5, 5, 5, 5, 6, 6, 6, 7, 7, 300
Mean = 5.45	Mean = 30.00
Median = 5.00	Median = 5.50
Mode = 5.00	Mode = 5.00
Standard Deviation = 1.04	Standard Deviation = 85.03

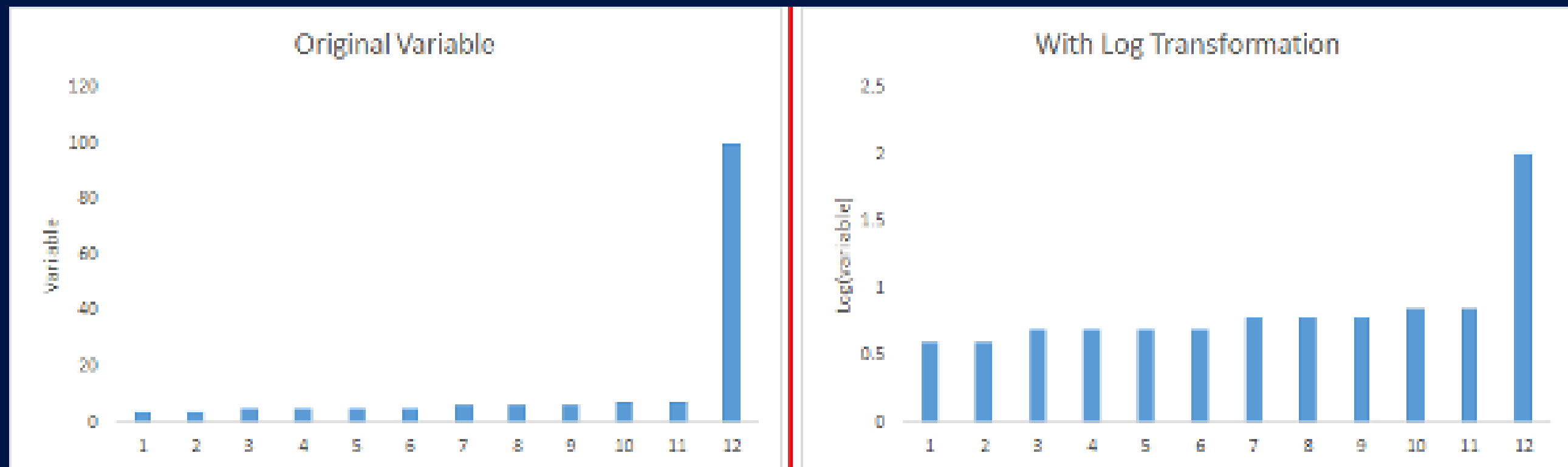
Variable Transformation

Transformation refers to the replacement of a variable by a function. It changes the **distribution's shape**.

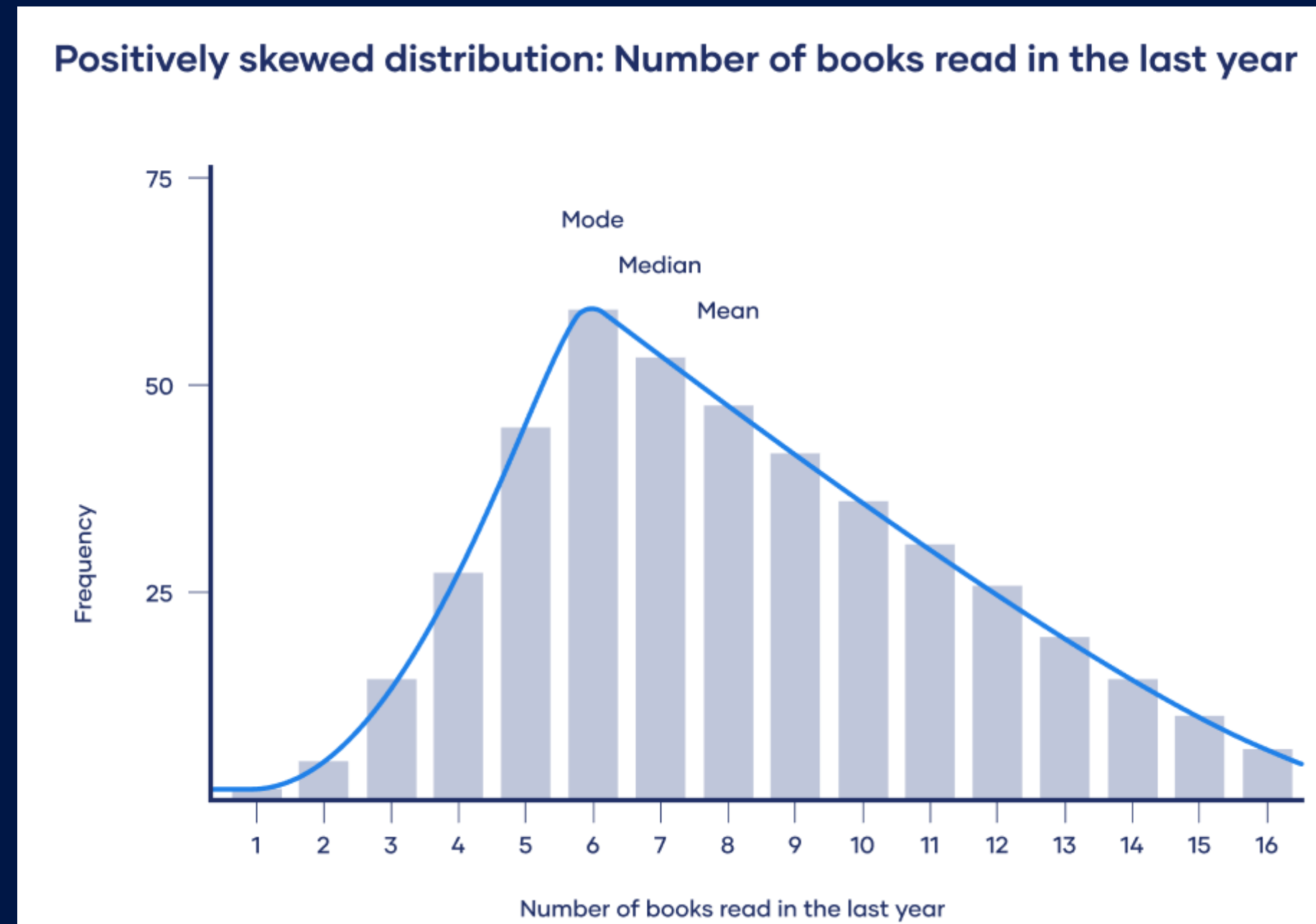
The goal of variable transformation is to make the data more normally distributed, stabilize variance, or linearize relationships. So, it is changing the actual nature of the data distribution.

For example, replacing a variable x with the **logarithm of x** , or **square/cube root** or **binning**, or **reciprocal** are transformations

1. Log Transformation: It is used to transform **complex non-linear** relationships into **linear relationships**. It changes the shape of the distribution on a distribution plot. It is generally used for reducing the **right skewness** of variables. However, It **can't be applied to zero or negative** values.



2. Square / Cube root: This transformation is used when **Symmetric distribution** is preferred over **skewed distribution**. For **right skewed** distribution, we take the **square/cube root** or **logarithm** of the variable and for **left skewed**, we take **square / cube** or **exponential** of variable.



3. Binning: It is used **to categorize** variables. It is performed on **original values, percentile, or frequency**. For example, we can categorize income into three categories, namely: **High Income, Average Income and Low Income**.

Binning is also done from an implementation point of view. For example, categorizing the sales agents into three **age group buckets of <30 years, 30-45 years and > 45 years**

We can also perform **co-variate binning** which depends on the value of **more than one** variable.

4. Centering the variable: The mean value of the variable is subtracted from each value of the variable.

centered variable value = variable value - mean of the variable

After **centering**, the **mean value** of the variable becomes **zero** but the R-squared and the residual **standard deviation** stay the same. So, it does **not affect the model's fit**. However, the **regression coefficient** of that variable **changes a lot**.

1 2 3 4 5 6 7 → mean 4

-3 -2 -1 0 1 2 3 → x-mean → mean 0

5. Standardizing the variable: This makes the mean of the variable 0 and the standard deviation 1.

Standardized variable = (variable value - mean of the variable) / standard deviation.

Variable transformation changes the distribution's shape whereas variable scaling changes the range of values.

Model Building

Model building is aimed at achieving not only **high accuracy on the training data** but also the ability to **generalize and perform** well on new, unseen data.

1. Data Preprocessing:

The prediction or classification accuracy of the model depends on the quality of the data available. To improve the quality of the data, the following preprocessing steps should be applied.

1. Variable Identification
2. Univariate Analysis
3. Bi-variate Analysis
4. Missing values treatment
5. Outlier treatment
6. Variable transformation
7. Variable creation

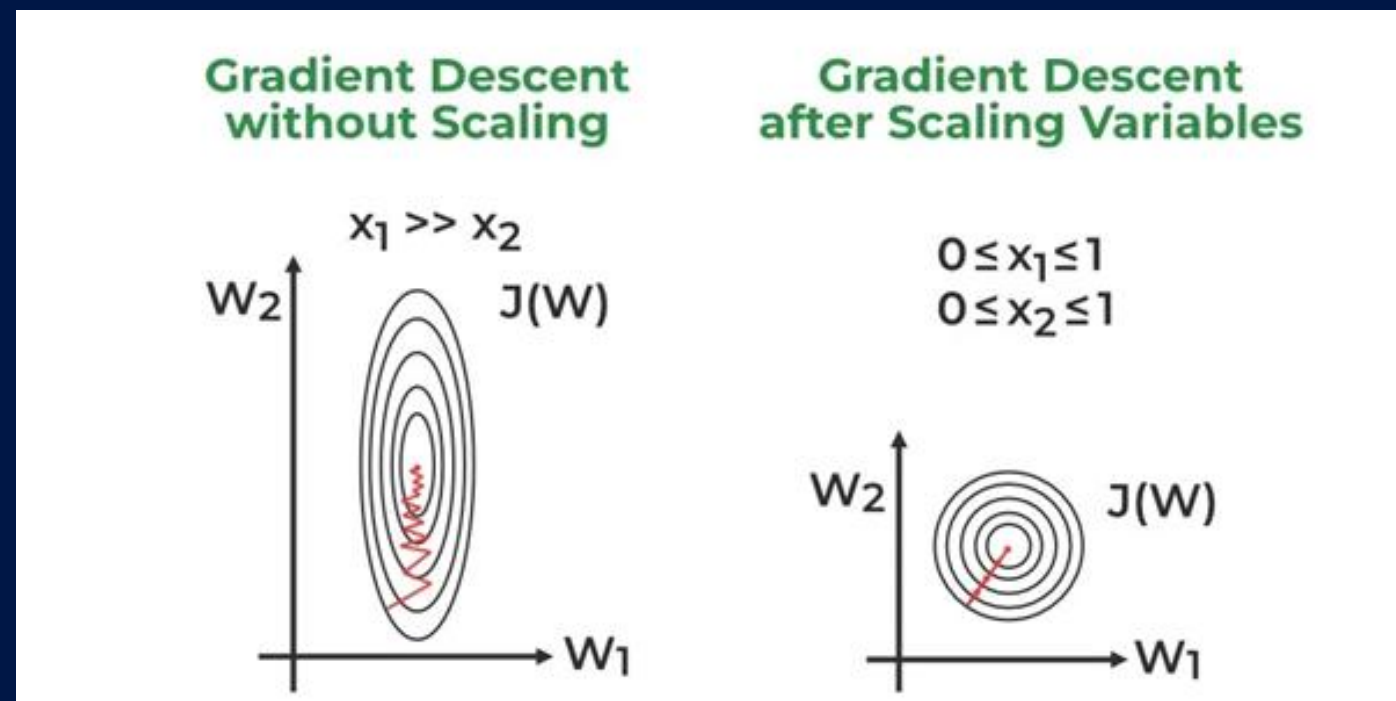
All these steps have been discussed already.

Model Building Contd. ...

2. Divide the data set: Based on the dataset quality and quantity of the data, one may choose to divide his dataset into three parts: **training and testing and validation data**. The sample Python code is given below. We have to import suitable packages for implementing each step.

```
from sklearn.model_selection import train_test_split  
train_data_x, test_data_x, train_data_y, test_data_y = train_test_split(data, y, test_size=0.25)
```

3. Scale the dataset: Suppose column **A** has data ranging from 1 to 1000 and column **B** has data ranging from 0 to 1. In that case column **A** can influence decision results provided the model even if it is not an important feature. But **after scaling** all the columns come in a **similar range**.



Some algorithms like K-nearest neighbors (**KNN**), use the **distance** between data points to make predictions in this case if the columns have different scales then the **distance can go higher**.

Model Building Contd. ...

4. Modeling the Data: Fit the data to the model.

The model should be selected based on the problem.

1. Whether our problem is a **regression problem** or a **classification problem**

2. Whether we want a model which is **more explainable** or we want a model which has a **higher accuracy**

For example, if the target value is **continuous**, consider it as a **regression** problem. A **decision tree** model can also be used in this case.

5. Evaluate the Model: After making the model we evaluate the model's accuracy. **Mean square error** is used for computing the accuracy of the model.

Explain the Bi-variate analysis of data.

Explain the methods used to detect and treat the missing values.

Quantitative and Qualitative Variables

Descriptive statistics in data science

T-Test

Chi-Square test

Correlation

PART-A Q&A

You can also download the updated Notes, PPTs and any material related to AIML subject from the Google Drive link

https://drive.google.com/drive/folders/1aG4DbOLy0yFVCTiOi2mh0O_t77kZfT6H?usp=sharing

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